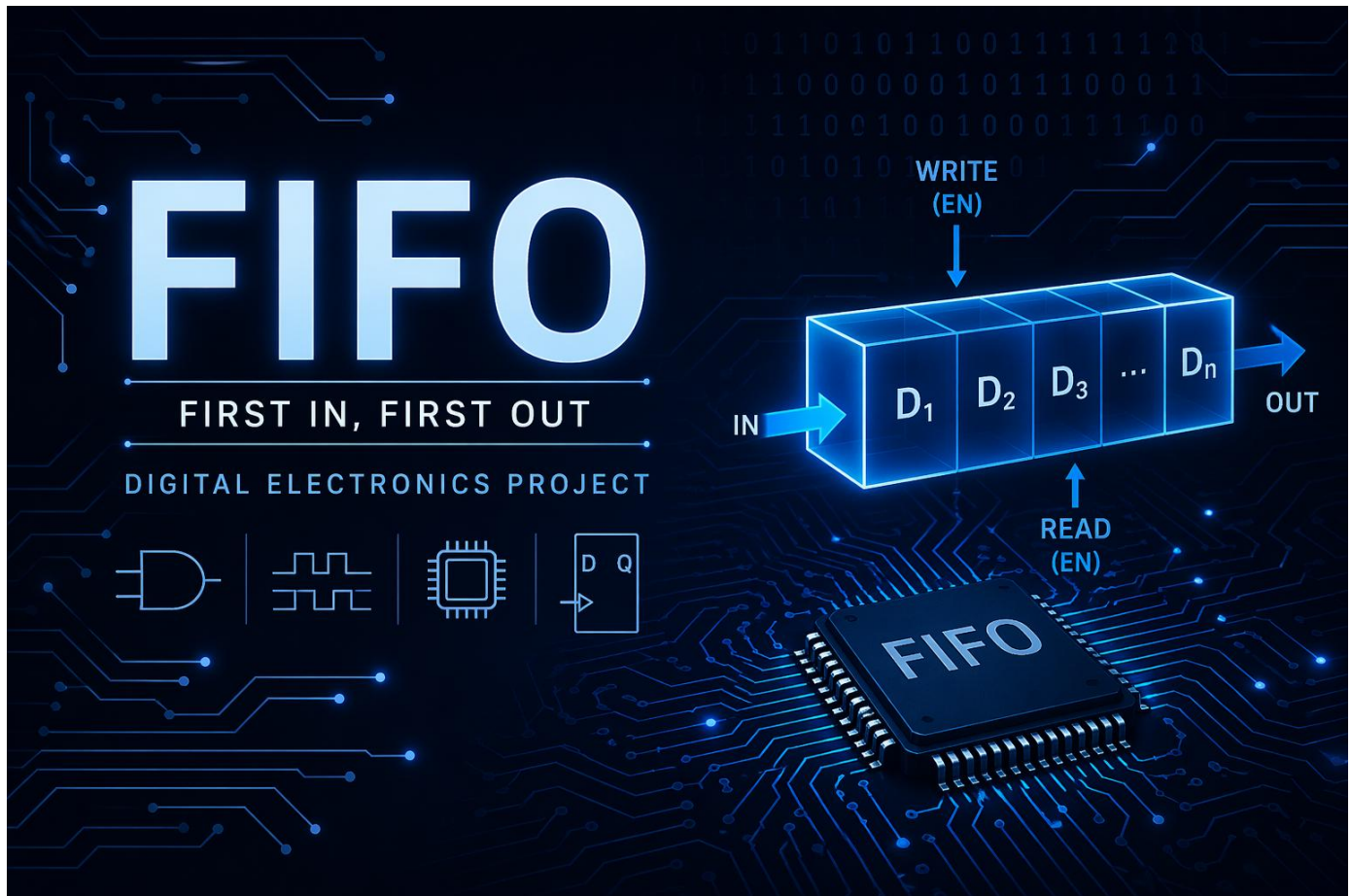


Synchronous FIFO Verification Project



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Verification plan :

	A	B	C	D	E
1	Label	Design Requirement Description	Stimulus Generation	Functional Coverage	Functionality Check
2	FIFO_1	During reset, all status signals (full, almost full, overflow, underflow) must be low, while empty should be asserted.	Apply reset at the beginning of simulation, then release it and randomized with 95% off of the simulation time	-	Immediate assertion along with reference model check to validate reset behavior
3	FIFO_2	After reset is released and write enable is active while FIFO is not full, input data should be written and write pointer should advance. Write acknowledge should be asserted.	Randomize input data and enable wr_en with a certain distribution during simulation.	Covers all combinations of wr_en and wr_ack	Concurrent assertion and reference model comparison to verify write operation
4	FIFO_3	When read enable is asserted and FIFO is not empty, the output should reflect the stored data at the current read pointer location.	Randomized read enable activation throughout simulation.	Covers all rd_en scenarios	Reference model checker ensures correctness of data_out
5	FIFO_4	When FIFO becomes full, the full flag must be asserted.	-	Covers all cases of full flag	Immediate assertion and reference model validation for full condition
6	FIFO_5	When FIFO is almost full (only one free location left), almostfull flag should be asserted.	-	Covers all almostfull cases	Immediate assertion and reference model validation
7	FIFO_6	When FIFO is empty, empty flag must be asserted.	-	Covers all empty conditions	Immediate assertion and reference model validation
8	FIFO_7	When FIFO has only one element left (almost empty), almostempty flag should be asserted.	-	Covers all almostempty conditions	Immediate assertion and reference model validation
9	FIFO_8	If FIFO is full and a write operation is attempted, overflow should be triggered.	-	Covers all overflow cases	Concurrent assertion and reference model validation
10	FIFO_9	If FIFO is empty and a read operation is attempted, underflow should be triggered.	-	Covers all underflow cases	Concurrent assertion and reference model validation
11	FIFO_10	When write enable is active and FIFO is not full, count and write pointer should increment.	-	-	Concurrent assertion to verify count and wr_ptr increment
12	FIFO_11	When read enable is active and FIFO is not empty, count should decrement and read pointer should increment.	-	-	Concurrent assertion to verify count and rd_ptr behavior

Interface :

```
interface FIFO_if(clk);
    parameter FIFO_WIDTH = 16;
    parameter FIFO_DEPTH = 8 ;
    input bit clk ;
    logic [FIFO_WIDTH-1:0] data_in;
    logic rst_n, wr_en, rd_en;
    logic [FIFO_WIDTH-1:0] data_out;
    logic wr_ack, overflow;
    logic full, empty, almostfull, almostempty, underflow;

    modport DUT (
        input clk ,rst_n,wr_en,rd_en,data_in,
        output data_out , wr_ack,overflow, full , empty, almostfull, almostempty, underflow
    );
    modport TEST (
        input clk ,data_out,wr_ack,overflow,full,empty,almostfull,almostempty,underflow,
        output rst_n, wr_en , rd_en,data_in
    );
    modport MONITOR (input clk, rst_n, wr_en, rd_en , data_in, data_out, wr_ack,
        overflow, full, empty, almostfull, almostempty, underflow);
endinterface
```

Design after bugs corrected:

```
module FIFO(FIFO_if.DUT fifoif);

localparam max_fifo_addr = $clog2(fifoif.FIFO_DEPTH);

reg [fifoif.FIFO_WIDTH-1:0] mem [fifoif.FIFO_DEPTH-1:0];

reg [max_fifo_addr-1:0] wr_ptr, rd_ptr;
reg [max_fifo_addr:0] count;

always @(posedge fifoif.clk or negedge fifoif.rst_n) begin

    if (! fifoif.rst_n) begin
        wr_ptr <= 0;
        fifoif.wr_ack <=0; //missing after reset
        fifoif.overflow <= 0; //missing after reset
    end
    else if (fifoif.wr_en && count < fifoif.FIFO_DEPTH) begin
        mem[wr_ptr] <= fifoif.data_in;
        fifoif.wr_ack <= 1;
        wr_ptr <= wr_ptr + 1;
        fifoif.overflow <= 0;
    end
    else begin
        fifoif.wr_ack <= 0;
        if (fifoif.full & fifoif.wr_en)
            fifoif.overflow <= 1;
        else
            fifoif.overflow <= 0;
    end
end

always @(posedge fifoif.clk or negedge fifoif.rst_n) begin
    if (! fifoif.rst_n) begin
        rd_ptr <= 0;
        fifoif.data_out <= 0; //missing after reset
        fifoif.underflow <= 0; //missing after reset
    end
    else if (fifoif.rd_en && count != 0) begin
        fifoif.data_out <= mem[rd_ptr];
        rd_ptr <= rd_ptr + 1;
        fifoif.underflow <= 0;
    end
    else begin
        if (fifoif.empty & fifoif.rd_en)
            fifoif.underflow <= 1;
        else
            fifoif.underflow <= 0;
    end
end

always @(posedge fifoif.clk or negedge fifoif.rst_n) begin
```

```

    if (!fifoif.rst_n) begin
        count <= 0;
    end
    else begin
        if ( ({fifoif.wr_en, fifoif.rd_en} == 2'b10) && ! fifoif.full)
            count <= count + 1;
        else if ( ({fifoif.wr_en, fifoif.rd_en} == 2'b01) && ! fifoif.empty)
            count <= count - 1;
        else if ( ({ fifoif.wr_en,  fifoif.rd_en} == 2'b11) && fifoif.full) //missing
cade
            count <= count - 1;
        else if ( ({ fifoif.wr_en,  fifoif.rd_en} == 2'b11) && count==0) //missing case
            count <= count + 1;
    end
end

assign fifoif.full = (count == fifoif.FIFO_DEPTH)? 1 : 0;
assign fifoif.empty = (count == 0)? 1 : 0;
assign fifoif.almostempty = (count == 1)? 1 : 0;
//assign fifoif.underflow = (fifoif.empty && fifoif.rd_en)? 1 : 0;    // its mentioned
in pdf its a sequential
assign fifoif.almostfull = (count == fifoif.FIFO_DEPTH-1)? 1 : 0;    // bug in -2


//assertions
`ifdef SIM

// RESET CHECK
always_comb begin
    if(!fifoif.rst_n) begin
        reset_state_ap: assert final(
            count == 0 &&
            rd_ptr == 0 &&
            wr_ptr == 0 &&
            fifoif.data_out == 0 &&
            fifoif.underflow == 0 &&
            fifoif.overflow == 0 &&
            fifoif.wr_ack == 0
        );

        reset_state_cp: cover final(
            count == 0 &&
            rd_ptr == 0 &&
            wr_ptr == 0 &&
            fifoif.data_out == 0 &&
            fifoif.underflow == 0 &&
            fifoif.overflow == 0 &&
            fifoif.wr_ack == 0
        );
    end
end
end

```

```

// FLAG CHECKS
always_comb begin
    if(count == fifoif.FIFO_DEPTH) begin
        full_flag_ap: assert final(fifoif.full == 1);
        full_flag_cp: cover final(fifoif.full == 1);
    end
end

always_comb begin
    if(count == 0) begin
        empty_flag_ap: assert final(fifoif.empty == 1);
        empty_flag_cp: cover final(fifoif.empty == 1);
    end
end

always_comb begin
    if(count == fifoif.FIFO_DEPTH-1) begin
        almost_full_flag_ap: assert final(fifoif.almostfull == 1);
        almost_full_flag_cp: cover final(fifoif.almostfull == 1);
    end
end

always_comb begin
    if(count == 1) begin
        almost_empty_flag_ap: assert final(fifoif.almostempty == 1);
        almost_empty_flag_cp: cover final(fifoif.almostempty == 1);
    end
end

// PROPERTIES

// Overflow behavior
property overflow_behavior_p;
    @(posedge fifoif.clk) disable iff (!fifoif.rst_n)
    (fifoif.full && fifoif.wr_en) | => (fifoif.overflow == 1);
endproperty

// Underflow behavior
property underflow_behavior_p;
    @(posedge fifoif.clk) disable iff (!fifoif.rst_n)
    (fifoif.empty && fifoif.rd_en) | => (fifoif.underflow == 1);
endproperty

// Write acknowledge behavior
property write_ack_behavior_p;
    @(posedge fifoif.clk) disable iff (!fifoif.rst_n)
    (count < fifoif.FIFO_DEPTH && fifoif.wr_en) | => (fifoif.wr_ack == 1);
endproperty

// Pointer boundary check
property pointer_limit_p;
    @(posedge fifoif.clk) disable iff (!fifoif.rst_n)

```

```

        (count <= fifoif.FIFO_DEPTH &&
         wr_ptr <= fifoif.FIFO_DEPTH &&
         rd_ptr <= fifoif.FIFO_DEPTH);
endproperty

// ASSERT
overflow_behavior_ap: assert property (overflow_behavior_p);
underflow_behavior_ap: assert property (underflow_behavior_p);
write_ack_behavior_ap: assert property (write_ack_behavior_p);
pointer_limit_ap: assert property (pointer_limit_p);

// COVER
overflow_behavior_cp: cover property (overflow_behavior_p);
underflow_behavior_cp: cover property (underflow_behavior_p);
write_ack_behavior_cp: cover property (write_ack_behavior_p);
pointer_limit_cp: cover property (pointer_limit_p);

`endif
endmodule

```

transaction pkg:

```

package FIFO_transaction_pkg;
parameter FIFO_WIDTH =16;
parameter FIFO_DEPTH =8;
class FIFO_transaction;
rand logic wr_en, rd_en, rst_n;
rand logic [FIFO_WIDTH-1:0] data_in;
logic [FIFO_WIDTH-1:0] data_out;
logic wr_ack, overflow;
logic full, empty, almostfull, almostempty, underflow;
int RD_EN_ON_DIST;
int WR_EN_ON_DIST;
constraint rst_n_dist { rst_n dist {1:=95, 0:=5};}
constraint wr_en_dist { wr_en dist {1:=WR_EN_ON_DIST, 0:=(100-WR_EN_ON_DIST)}};
constraint rd_en_dist { rd_en dist {1:=RD_EN_ON_DIST, 0:=(100-RD_EN_ON_DIST)}};

function new(int RD_EN_ON_DIST=30, int WR_EN_ON_DIST = 70);
this.RD_EN_ON_DIST= RD_EN_ON_DIST;
this.WR_EN_ON_DIST= WR_EN_ON_DIST;
endfunction
endclass
endpackage

```

coverage pkg:

```
package FIFO_coverage_pkg;
import FIFO_transaction_pkg::* ;
class FIFO_coverage;
FIFO_transaction F_cvg_txn = new();

covergroup FIFOcg ;

    wr_en_cp: coverpoint F_cvg_txn.wr_en;
    rd_en_cp: coverpoint F_cvg_txn.rd_en;
    full_cp: coverpoint F_cvg_txn.full;
    almostfull_cp: coverpoint F_cvg_txn.almostfull;
    empty_cp: coverpoint F_cvg_txn.empty;
    almostempty_cp: coverpoint F_cvg_txn.almostempty;
    overflow_cp: coverpoint F_cvg_txn.overflow;
    underflow_cp: coverpoint F_cvg_txn.underflow;
    wr_ack_cp: coverpoint F_cvg_txn.wr_ack;

    full_cross: cross wr_en_cp, rd_en_cp, full_cp{
        ignore_bins rd_full = binsof(rd_en_cp)intersect{1} &&
        binsof(full_cp)intersect{1}
    ;}
    almostfull_cross: cross wr_en_cp, rd_en_cp, almostfull_cp;
    empty_cross: cross wr_en_cp, rd_en_cp, empty_cp;
    almostempty_cross: cross wr_en_cp, rd_en_cp, almostempty_cp;

    overflow_cross: cross wr_en_cp, rd_en_cp, overflow_cp{
        ignore_bins not_wr_overflow = binsof(wr_en_cp)intersect{0} &&
        binsof(overflow_cp)intersect{1}
    ;}

    underflow_cross: cross wr_en_cp, rd_en_cp, underflow_cp{
        ignore_bins not_rd_overflow = binsof(rd_en_cp)intersect{0} &&
        binsof(underflow_cp)intersect{1}
    ;}

    wr_ack_cross: cross wr_en_cp, rd_en_cp, wr_ack_cp{
        ignore_bins not_wren_wrack = binsof(wr_en_cp)intersect{0} &&
        binsof(wr_ack_cp)intersect{1}
    ;}

endgroup

function new();
    FIFOcg =new();
endfunction

function void sample_data(input FIFO_transaction F_txn);
    F_cvg_txn=F_txn;
    FIFOcg.sample();
endfunction
```

```
endfunction
endclass
endpackage
```

shared pkg:

```
package shared_pkg;
int correct_count=0;
int error_count=0;
bit test_finished;
event trigger;
endpackage
```

scoreboard pkg:

```
package FIFO_scoreboard_pkg;
import FIFO_transaction_pkg::* ;
import shared_pkg::* ;
class FIFO_scoreboard;
localparam max_fifo_addr = $clog2(FIFO_DEPTH);
logic [FIFO_WIDTH-1:0] data_out_ref;
logic full_ref, almostfull_ref, empty_ref, almostempty_ref, overflow_ref, underflow_ref,
wr_ack_ref ;
reg [FIFO_WIDTH-1:0] mem [FIFO_DEPTH-1:0];
reg [max_fifo_addr-1:0] wr_ptr, rd_ptr;
reg [max_fifo_addr:0] count;

//Check data
function void check_data(input FIFO_transaction fifotrans_obj_checkdata);
    reference_model(fifotrans_obj_checkdata);
    if (data_out_ref == fifotrans_obj_checkdata.data_out && full_ref ==
fifotrans_obj_checkdata.full && almostfull_ref == fifotrans_obj_checkdata.almostfull &&
empty_ref == fifotrans_obj_checkdata.empty && almostempty_ref ==
fifotrans_obj_checkdata.almostempty && overflow_ref == fifotrans_obj_checkdata.overflow
&& underflow_ref == fifotrans_obj_checkdata.underflow && wr_ack_ref ==
fifotrans_obj_checkdata.wr_ack ) begin
        correct_count=correct_count+1;
        $display("Correct in FIFO");
    end
    else begin
        error_count=error_count+1;
        $display("Error in FIFO");
        $display("rst_n=%0d",fifotrans_obj_checkdata.rst_n);
        $display("data_in=%0d",fifotrans_obj_checkdata.data_in);
        $display("data_out=%0d",fifotrans_obj_checkdata.data_out);
        $display("data_out_ref=%0d",data_out_ref);
        $display("wr_ack=%0d",fifotrans_obj_checkdata.wr_ack);
        $display("wr_ack_ref=%0d",wr_ack_ref);
        $display("overflow=%0d",fifotrans_obj_checkdata.overflow);
        $display("overflow_ref=%0d",overflow_ref);
        $display("full=%0d",fifotrans_obj_checkdata.full);
    end
end
endclass
```



```

    $display("full_ref=%0d",full_ref);
    $display("empty=%0d",fifotrans_obj_checkdata.empty);
    $display("empty_ref=%0d",empty_ref);
    $display("almostfull=%0d",fifotrans_obj_checkdata.almostfull);
    $display("almostfull_ref=%0d",almostfull_ref);
    $display("almostempty=%0d",fifotrans_obj_checkdata.almostempty);
    $display("almostempty_ref=%0d",almostempty_ref);
    $display("underflow=%0d",fifotrans_obj_checkdata.underflow);
    $display("underflow_ref=%0d",underflow_ref);
    $display("wr_en=%0d",fifotrans_obj_checkdata.wr_en);
    $display("rd_en=%0d",fifotrans_obj_checkdata.rd_en);
    $display("count=%0d",count);
end
endfunction

//Reference model
function void reference_model(input FIFO_transaction fifotrans_obj_refmodel);
//Write data
    if (! fifotrans_obj_refmodel.rst_n) begin
        wr_ptr = 0;
        wr_ack_ref = 0;
        overflow_ref = 0;
    end

    else if (fifotrans_obj_refmodel.wr_en && count < FIFO_DEPTH) begin
        mem[wr_ptr] = fifotrans_obj_refmodel.data_in;
        wr_ack_ref = 1;
        wr_ptr = wr_ptr + 1;
        overflow_ref = 0;
    end

    else begin
        wr_ack_ref = 0;
        if (full_ref && fifotrans_obj_refmodel.wr_en)
            overflow_ref = 1;
        else
            overflow_ref = 0;
    end

//Read data
    if (!fifotrans_obj_refmodel.rst_n) begin
        rd_ptr = 0;
        underflow_ref = 0;
        data_out_ref = 0;
    end

    else if (fifotrans_obj_refmodel.rd_en && !empty_ref) begin
        data_out_ref = mem[rd_ptr];
        rd_ptr = rd_ptr + 1;
        underflow_ref = 0;
    end

    else begin

```

```

        if ( fifotrans_obj_refmodel.rd_en && empty_ref)
            underflow_ref = 1;
        else
            underflow_ref = 0;
        end
    //Handle count
    if (!fifotrans_obj_refmodel.rst_n) begin
        count = 0;
    end
    else begin
        if ( ({fifotrans_obj_refmodel.wr_en, fifotrans_obj_refmodel.rd_en} == 2'b10) &&
! full_ref)
            count = count + 1;
        else if ( ({fifotrans_obj_refmodel.wr_en, fifotrans_obj_refmodel.rd_en} == 2'b01)
&& ! empty_ref)
            count = count - 1;
        else if ( ({ fifotrans_obj_refmodel.wr_en,  fifotrans_obj_refmodel.rd_en} ==
2'b11) && full_ref) //read ->decrement
            count = count - 1;
        else if ( ({ fifotrans_obj_refmodel.wr_en,  fifotrans_obj_refmodel.rd_en} ==
2'b11) && empty_ref) //write ->increment
            count = count + 1;
        end
    //Handle flags
    full_ref = (count == FIFO_DEPTH)? 1 : 0;
    empty_ref = (count == 0)? 1 : 0;
    almostfull_ref = (count == FIFO_DEPTH-1)? 1 : 0;
    almostempty_ref = (count == 1)? 1 : 0;
endfunction

endclass

endpackage

```

testbench:

```

import shared_pkg::* ;
import FIFO_transaction_pkg::* ;
import FIFO_scoreboard_pkg::* ;
import FIFO_coverage_pkg::* ;
module FIFO_tb(FIFO_if.TEST fifoif);
FIFO_transaction obj_transs ;
initial begin
obj_transs=new();
    fifoif.rst_n=0;
    fifoif.wr_en=0;
    fifoif.rd_en=0;
    fifoif.data_in=0;
    -> trigger;
    @(negedge fifoif.clk);
    fifoif.rst_n=1;

```

```

repeat(10000)
begin
    assert(obj_transss.randomize());
    fifoif.rst_n = obj_transss.rst_n;
    fifoif.data_in = obj_transss.data_in;
    fifoif.wr_en = obj_transss.wr_en;
    fifoif.rd_en = obj_transss.rd_en;
    -> trigger;
    @(negedge fifoif.clk);
end

    test_finished=1;
-> trigger;
end

endmodule

```

top module :

```

module FIFO_top();

bit clk;
    initial begin
        clk=0;
        forever #1 clk=~clk;
    end
FIFO_if fifoif (clk);
FIFO DUT (fifoif);
FIFO_tb TEST (fifoif);
FIFO_monitor MONITOR (fifoif);
endmodule

```

monitor :

```

import shared_pkg::* ;
import FIFO_transaction_pkg::* ;
import FIFO_scoreboard_pkg::* ;
import FIFO_coverage_pkg::* ;
module FIFO_monitor(FIFO_if.MONITOR fifoif);
FIFO_transaction obj_trans=new();
FIFO_coverage obj_cvr=new();
FIFO_scoreboard obj_sb=new();

initial begin

forever begin
    wait(trigger.triggered);
    @(negedge fifoif.clk);
    obj_trans.rst_n = fifoif.rst_n;
    obj_trans.wr_en = fifoif.wr_en;
    obj_trans.rd_en = fifoif.rd_en;
    obj_trans.data_in = fifoif.data_in;

```

```

obj_trans.overflow = fifoif.overflow;
obj_trans.underflow = fifoif.underflow;
obj_trans.empty = fifoif.empty;
obj_trans.almostempty = fifoif.almostempty;
obj_trans.full = fifoif.full;
obj_trans.almostfull = fifoif.almostfull;
obj_trans.data_out = fifoif.data_out;
obj_trans.wr_ack = fifoif.wr_ack;

fork
//first
begin
obj_cvr.sample_data(obj_trans);
end

//second
begin
obj_sb.check_data(obj_trans);
end
join

if (test_finished==1) begin
    $display("Correct count = %d", correct_count);
    $display("Error count = %d", error_count);
    $stop;
end
end
end
endmodule

```

do file :

```

1  vlib work
2  vlog +define+SIM -f src_files.list +cover -covercells
3  vsim -voptargs=+acc work.FIFO_top -cover
4
5  add wave *
6  run 0
7  add wave -position insertpoint sim:/FIFO_top/fifoif/*
8  add wave -position insertpoint sim:/FIFO_top/DUT/*
9  coverage save fifo.ucdb -onexit
10 run -all |

```

List:

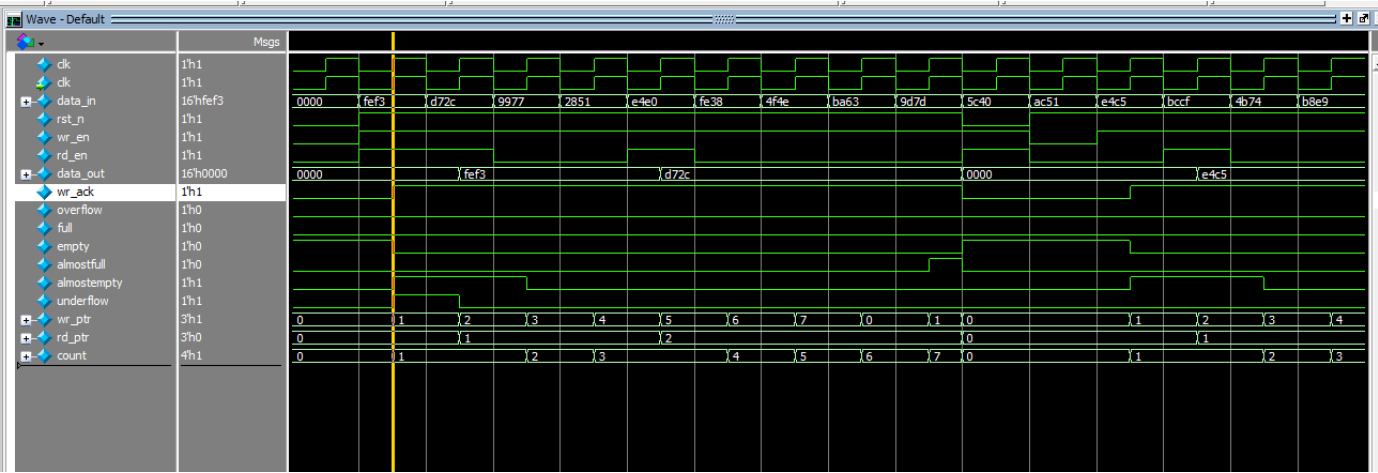
```

1  FIFO.sv
2  FIFO_if.sv
3  transaction_pkg.sv
4  shared_pkg.sv
5  scoreboard_pkg.sv
6  coverage_pkg.sv
7  monitor.sv
8  FIFO_tb.sv
9  FIFO_top.sv

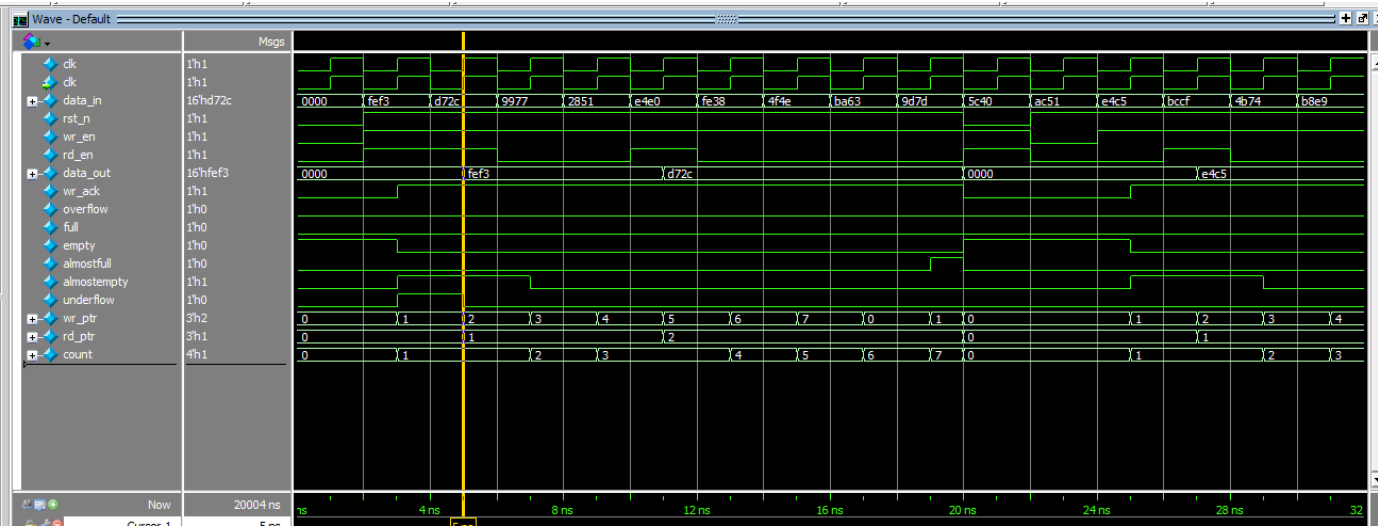
```

Questasim waveform :

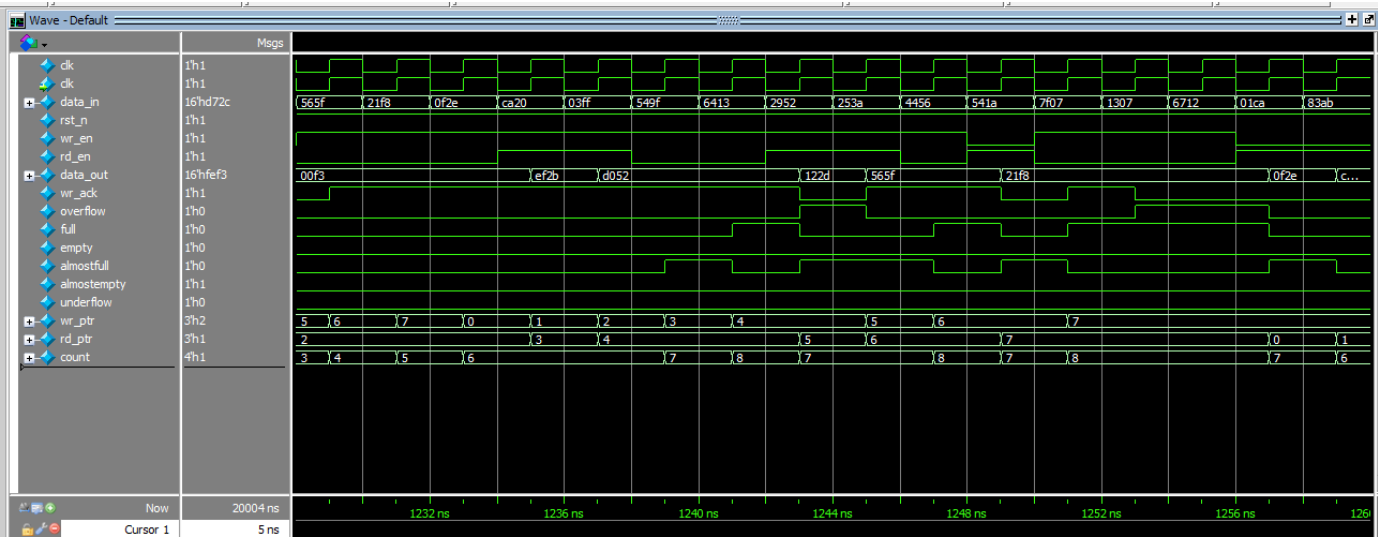
First in :



First out :



Another snippet for the waveform :



Transcript showing zero error count :

```
# Correct count =      10002
# Error count =        0
# ** Note: $stop      : monitor.sv(43)
#   Time: 20004 ns   Iteration: 1   Instance: /FIFO_top/MONITOR
# Break in Module FIFO_monitor at monitor.sv line 43

VSIM(paused)>
```

Coverage snippets from Questasim :

Cover directives

Cover Directives

Name	Language	Enabled	Log	Count	AtLeast	Limit	Weight	Cmplt %	Cmplt graph	Incl
 /FIFO_top/DUT/reset_cvr	SVA		Off	460	1	Unli...	1	100%		
 /FIFO_top/DUT/full_cvr	SVA		Off	830	1	Unli...	1	100%		
 /FIFO_top/DUT/empty_cvr	SVA		Off	545	1	Unli...	1	100%		
 /FIFO_top/DUT/almostfull_cvr	SVA		Off	1062	1	Unli...	1	100%		
 /FIFO_top/DUT/almostempty_cvr	SVA		Off	627	1	Unli...	1	100%		
 /FIFO_top/DUT/overflow_cvr	SVA		Off	1621	1	Unli...	1	100%		
 /FIFO_top/DUT/underflow_cvr	SVA		Off	237	1	Unli...	1	100%		
 /FIFO_top/DUT/wr_ack_cvr	SVA		Off	4806	1	Unli...	1	100%		
 /FIFO_top/DUT/threshold_cvr	SVA		Off	9521	1	Unli...	1	100%		

Analysis Cover Directives Covergroups Wave Assertions monitor.sv

Functional coverage :

Covergroups									
Name	Class Type	Coverage	Goal	% of Goal	Status	Included	Merge_instances	Get_inst_coverage	Comment
/FIFO_coverage_p...		100.00%							
TYPE FIFOcg		100.00%	100	100.00...		✓	auto(1)		
+ CVP FIFOc...		100.00%	100	100.00...		✓			
+ CVP FIFOc...		100.00%	100	100.00...		✓			
+ CVP FIFOc...		100.00%	100	100.00...		✓			
+ CVP FIFOc...		100.00%	100	100.00...		✓			
+ CVP FIFOc...		100.00%	100	100.00...		✓			
+ CVP FIFOc...		100.00%	100	100.00...		✓			
+ CVP FIFOc...		100.00%	100	100.00...		✓			
+ CVP FIFOc...		100.00%	100	100.00...		✓			
+ CROSS FIF...		100.00%	100	100.00...		✓			
+ CROSS FIF...		100.00%	100	100.00...		✓			
+ CROSS FIF...		100.00%	100	100.00...		✓			
+ CROSS FIF...		100.00%	100	100.00...		✓			
+ CROSS FIF...		100.00%	100	100.00...		✓			
+ CROSS FIF...		100.00%	100	100.00...		✓			
+ CROSS FIF...		100.00%	100	100.00...		✓			
+ CROSS FIF...		100.00%	100	100.00...		✓			

Analysis Cover Directives Covergroups Wave Assertions monitor.sv

Assertion coverage

Assertions							+	
Name	Assertion Type	Language	Enable	Failure Count	Pass Count	Active Count		
+ /FIFO_top/DUT/reset_assert	Immediate	SVA	on	0	1	-		
/FIFO_top/DUT/full_check	Immediate	SVA	on	0	1	-		
/FIFO_top/DUT/empty_check	Immediate	SVA	on	0	1	-		
/FIFO_top/DUT/almostfull_check	Immediate	SVA	on	0	1	-		
/FIFO_top/DUT/almostempty_check	Immediate	SVA	on	0	1	-		
+ /FIFO_top/DUT/overflow_check	Concurrent	SVA	on	0	1	-		
+ /FIFO_top/DUT/underflow_check	Concurrent	SVA	on	0	1	-		
+ /FIFO_top/DUT/wr_ack_check	Concurrent	SVA	on	0	1	-		
+ /FIFO_top/DUT/threshold_check	Concurrent	SVA	on	0	1	-		
/FIFO_top/TEST/#ublk#182146786#18/immed__19	Immediate	SVA	on	0	1	-		

Analysis x Cover Directives x Covergroups x Wave x Assertions x monitor.sv x

Coverage report :

Coverage Report by instance with details

=====
=== Instance: /FIFO_top/fifoif
=== Design Unit: work.FIFO_if
=====

Toggle Coverage:

Enabled Coverage	Bins	Hits	Misses	Coverage
-----	----	----	-----	-----
Toggles	86	86	0	100.00%

=====Toggle Details=====

Toggle Coverage for instance /FIFO_top/fifoif --

"Coverage"	Node	1H->0L	0L->1H
---	-----		
	almostempty	1	1
100.00			
	almostfull	1	1
100.00			
	clk	1	1
100.00			
	data_in[15-0]	1	1
100.00			
	data_out[15-0]	1	1
100.00			
	empty	1	1
100.00			
	full	1	1
100.00			
	overflow	1	1
100.00			
	rd_en	1	1
100.00			
	rst_n	1	1
100.00			
	underflow	1	1
100.00			
	wr_ack	1	1
100.00			
	wr_en	1	1
100.00			

Total Node Count = 43
Toggled Node Count = 43
Untoggled Node Count = 0

Toggle Coverage = 100.00% (86 of 86 bins)

=====
=== Instance: /FIFO_top/DUT
=== Design Unit: work.FIFO
=====

Assertion Coverage:

Assertions	9	9	0	100.00%

Name	File(Line)	Failure	Pass	

	Count	Count
/FIFO_top/DUT/reset_state_ap		
FIFO.sv(82)	0	1
/FIFO_top/DUT/full_flag_ap		
FIFO.sv(107)	0	1
/FIFO_top/DUT/empty_flag_ap		
FIFO.sv(114)	0	1
/FIFO_top/DUT/almost_full_flag_ap		
FIFO.sv(121)	0	1
/FIFO_top/DUT/almost_empty_flag_ap		
FIFO.sv(128)	0	1
/FIFO_top/DUT/overflow_behavior_ap		
FIFO.sv(164)	0	1
/FIFO_top/DUT/underflow_behavior_ap		
FIFO.sv(165)	0	1
/FIFO_top/DUT/write_ack_behavior_ap		
FIFO.sv(166)	0	1
/FIFO_top/DUT/pointer_limit_ap		
FIFO.sv(167)	0	1

Branch Coverage:

Enabled Coverage	Bins	Hits	Misses	Coverage
-----	----	----	-----	-----
Branches	35	35	0	100.00%

=====Branch Details=====

Branch Coverage for instance /FIFO_top/DUT

Line	Item	Count	Source
----	----	-----	-----
File FIFO.sv			
-----IF Branch-----			
12		10462	Count coming in to IF
12	1	941	if (! fifoif.rst_n)
begin			
17	1	5044	else if (fifoif.wr_en &&
count < fifoif.FIFO_DEPTH) begin			
23	1	4477	else begin
Branch totals: 3 hits of 3 branches = 100.00%			
-----IF Branch-----			
25		4477	Count coming in to IF
25	1	1712	if (fifoif.full &
fifoif.wr_en)			
27	1	2765	else
Branch totals: 2 hits of 2 branches = 100.00%			
-----IF Branch-----			
33		10462	Count coming in to IF
33	1	941	if (! fifoif.rst_n)
begin			
38	1	2623	else if (fifoif.rd_en &&
count != 0) begin			
43	1	6898	else begin
Branch totals: 3 hits of 3 branches = 100.00%			
-----IF Branch-----			
44		6898	Count coming in to IF
44	1	252	if (fifoif.empty &
fifoif.rd_en)			
46	1	6646	else
Branch totals: 2 hits of 2 branches = 100.00%			

```

-----IF Branch-----
52                                9205    Count coming in to IF
52                                934      if (!fifoif.rst_n) begin
55                                8271    else begin
Branch totals: 2 hits of 2 branches = 100.00%

-----IF Branch-----
56                                8271    Count coming in to IF
56                                3509    if (
({fifoif.wr_en, fifoif.rd_en} == 2'b10) && ! fifoif.full)
58                                766      else if (
({fifoif.wr_en, fifoif.rd_en} == 2'b01) && ! fifoif.empty)
60                                502      else if ( ({
fifoif.wr_en,  fifoif.rd_en} == 2'b11) && fifoif.full) //missing cade
62                                180      else if ( ({
fifoif.wr_en,  fifoif.rd_en} == 2'b11) && count==0) //missing case
3314    All False Count
Branch totals: 5 hits of 5 branches = 100.00%

-----IF Branch-----
67                                5405    Count coming in to IF
67                                830      assign fifoif.full = (count
== fifoif.FIFO_DEPTH)? 1 : 0;
67                                4575    assign fifoif.full = (count
== fifoif.FIFO_DEPTH)? 1 : 0;
Branch totals: 2 hits of 2 branches = 100.00%

-----IF Branch-----
68                                5405    Count coming in to IF
68                                545      assign fifoif.empty = (count
== 0)? 1 : 0;
68                                4860    assign fifoif.empty = (count
== 0)? 1 : 0;
Branch totals: 2 hits of 2 branches = 100.00%

-----IF Branch-----
69                                5405    Count coming in to IF
69                                627      assign fifoif.almostempty =
(count == 1)? 1 : 0;
69                                4778    assign fifoif.almostempty =
(count == 1)? 1 : 0;
Branch totals: 2 hits of 2 branches = 100.00%

-----IF Branch-----
71                                5405    Count coming in to IF
71                                1062    assign fifoif.almostfull =
(count == fifoif.FIFO_DEPTH-1)? 1 : 0; // bug in -2
71                                4343    assign fifoif.almostfull =
(count == fifoif.FIFO_DEPTH-1)? 1 : 0; // bug in -2
Branch totals: 2 hits of 2 branches = 100.00%

-----IF Branch-----
80                                9692    Count coming in to IF
80                                918      if(!fifoif.rst_n) begin
8774    All False Count
Branch totals: 2 hits of 2 branches = 100.00%

-----IF Branch-----
106                               5406    Count coming in to IF
106                               830      if(count ==
fifoif.FIFO_DEPTH) begin
4576    All False Count

```

Branch totals: 2 hits of 2 branches = 100.00%

```
-----IF Branch-----
113                               5406    Count coming in to IF
113                1              545      if(count == 0) begin
                               4861      All False Count
```

Branch totals: 2 hits of 2 branches = 100.00%

```
-----IF Branch-----
120                               5406    Count coming in to IF
120                1              1062      if(count ==
fifoif.FIFO_DEPTH-1) begin
                               4344      All False Count
```

Branch totals: 2 hits of 2 branches = 100.00%

```
-----IF Branch-----
127                               5406    Count coming in to IF
127                1              627      if(count == 1) begin
                               4779      All False Count
```

Branch totals: 2 hits of 2 branches = 100.00%

Condition Coverage:

Enabled Coverage	Bins	Covered	Misses	Coverage
-----	----	----	-----	-----
Conditions	28	26	2	92.85%

=====Condition Details=====

Condition Coverage for instance /FIFO_top/DUT --

File FIFO.sv

-----Focused Condition View-----

Line 17 Item 1 (fifoif.wr_en && (count < fifoif.FIFO_DEPTH))

Condition totals: 2 of 2 input terms covered = 100.00%

Input Term	Covered	Reason for no coverage	Hint
-----	-----	-----	-----
fifoif.wr_en	Y		
(count < fifoif.FIFO_DEPTH)	Y		

Rows:	Hits	FEC Target	Non-masking condition(s)
-----	-----	-----	-----
Row 1:	1	fifoif.wr_en_0	-
Row 2:	1	fifoif.wr_en_1	(count < fifoif.FIFO_DEPTH)
Row 3:	1	(count < fifoif.FIFO_DEPTH)_0	fifoif.wr_en
Row 4:	1	(count < fifoif.FIFO_DEPTH)_1	fifoif.wr_en

-----Focused Condition View-----

Line 25 Item 1 (fifoif.full & fifoif.wr_en)

Condition totals: 1 of 2 input terms covered = 50.00%

Input Term	Covered	Reason for no coverage	Hint
-----	-----	-----	-----
fifoif.full	N	'_0' not hit	Hit '_0'
fifoif.wr_en	Y		

Rows:	Hits	FEC Target	Non-masking condition(s)
-----	-----	-----	-----
Row 1:	***0***	fifoif.full_0	fifoif.wr_en
Row 2:	1	fifoif.full_1	fifoif.wr_en
Row 3:	1	fifoif.wr_en_0	fifoif.full
Row 4:	1	fifoif.wr_en_1	fifoif.full

-----Focused Condition View-----

Line 38 Item 1 (fifoif.rd_en && (count != 0))

Condition totals: 2 of 2 input terms covered = 100.00%

Input Term	Covered	Reason for no coverage	Hint
-----	-----	-----	-----
fifoif.rd_en	Y		
(count != 0)	Y		

Rows:	Hits	FEC Target	Non-masking condition(s)
-----	-----	-----	-----
Row 1:	1	fifoif.rd_en_0	-
Row 2:	1	fifoif.rd_en_1	(count != 0)
Row 3:	1	(count != 0)_0	fifoif.rd_en
Row 4:	1	(count != 0)_1	fifoif.rd_en

-----Focused Condition View-----

Line 44 Item 1 (fifoif.empty & fifoif.rd_en)

Condition totals: 1 of 2 input terms covered = 50.00%

Input Term	Covered	Reason for no coverage	Hint
-----	-----	-----	-----
fifoif.empty	N	'_0' not hit	Hit '_0'
fifoif.rd_en	Y		

Rows:	Hits	FEC Target	Non-masking condition(s)
-----	-----	-----	-----
Row 1:	***0***	fifoif.empty_0	fifoif.rd_en
Row 2:	1	fifoif.empty_1	fifoif.rd_en
Row 3:	1	fifoif.rd_en_0	fifoif.empty
Row 4:	1	fifoif.rd_en_1	fifoif.empty

-----Focused Condition View-----

Line 56 Item 1 ((~fifoif.rd_en && fifoif.wr_en) && ~fifoif.full)

Condition totals: 3 of 3 input terms covered = 100.00%

Input Term	Covered	Reason for no coverage	Hint
-----	-----	-----	-----
fifoif.rd_en	Y		
fifoif.wr_en	Y		
fifoif.full	Y		

Rows:	Hits	FEC Target	Non-masking condition(s)
-----	-----	-----	-----
Row 1:	1	fifoif.rd_en_0	(~fifoif.full && fifoif.wr_en)
Row 2:	1	fifoif.rd_en_1	-
Row 3:	1	fifoif.wr_en_0	~fifoif.rd_en
Row 4:	1	fifoif.wr_en_1	(~fifoif.full && ~fifoif.rd_en)
Row 5:	1	fifoif.full_0	(~fifoif.rd_en && fifoif.wr_en)
Row 6:	1	fifoif.full_1	(~fifoif.rd_en && fifoif.wr_en)

-----Focused Condition View-----

Line 58 Item 1 ((fifoif.rd_en && ~fifoif.wr_en) && ~fifoif.empty)

Condition totals: 3 of 3 input terms covered = 100.00%

Input Term	Covered	Reason for no coverage	Hint
-----	-----	-----	-----
fifoif.rd_en	Y		
fifoif.wr_en	Y		
fifoif.empty	Y		

Rows:	Hits	FEC Target	Non-masking condition(s)
-------	------	------------	--------------------------

Row 1:	1	fifoif.rd_en_0	-
Row 2:	1	fifoif.rd_en_1	(~fifoif.empty && ~fifoif.wr_en)
Row 3:	1	fifoif.wr_en_0	(~fifoif.empty && fifoif.rd_en)
Row 4:	1	fifoif.wr_en_1	fifoif.rd_en
Row 5:	1	fifoif.empty_0	(fifoif.rd_en && ~fifoif.wr_en)
Row 6:	1	fifoif.empty_1	(fifoif.rd_en && ~fifoif.wr_en)

-----Focused Condition View-----

Line 60 Item 1 ((fifoif.rd_en && fifoif.wr_en) && fifoif.full)
Condition totals: 3 of 3 input terms covered = 100.00%

Input Term	Covered	Reason for no coverage	Hint
fifoif.rd_en	Y		
fifoif.wr_en	Y		
fifoif.full	Y		

Rows:	Hits	FEC Target	Non-masking condition(s)
Row 1:	1	fifoif.rd_en_0	-
Row 2:	1	fifoif.rd_en_1	(fifoif.full && fifoif.wr_en)
Row 3:	1	fifoif.wr_en_0	fifoif.rd_en
Row 4:	1	fifoif.wr_en_1	(fifoif.full && fifoif.rd_en)
Row 5:	1	fifoif.full_0	(fifoif.rd_en && fifoif.wr_en)
Row 6:	1	fifoif.full_1	(fifoif.rd_en && fifoif.wr_en)

-----Focused Condition View-----

Line 62 Item 1 ((fifoif.rd_en && fifoif.wr_en) && (count == 0))
Condition totals: 3 of 3 input terms covered = 100.00%

Input Term	Covered	Reason for no coverage	Hint
fifoif.rd_en	Y		
fifoif.wr_en	Y		
(count == 0)	Y		

Rows:	Hits	FEC Target	Non-masking condition(s)
Row 1:	1	fifoif.rd_en_0	-
Row 2:	1	fifoif.rd_en_1	((count == 0) && fifoif.wr_en)
Row 3:	1	fifoif.wr_en_0	fifoif.rd_en
Row 4:	1	fifoif.wr_en_1	((count == 0) && fifoif.rd_en)
Row 5:	1	(count == 0)_0	(fifoif.rd_en && fifoif.wr_en)
Row 6:	1	(count == 0)_1	(fifoif.rd_en && fifoif.wr_en)

-----Focused Condition View-----

Line 67 Item 1 (count == fifoif.FIFO_DEPTH)
Condition totals: 1 of 1 input term covered = 100.00%

Input Term	Covered	Reason for no coverage	Hint
(count == fifoif.FIFO_DEPTH)	Y		

Rows:	Hits	FEC Target	Non-masking condition(s)
Row 1:	1	(count == fifoif.FIFO_DEPTH)_0	-
Row 2:	1	(count == fifoif.FIFO_DEPTH)_1	-

-----Focused Condition View-----

Line 68 Item 1 (count == 0)
Condition totals: 1 of 1 input term covered = 100.00%

Input Term		Covered	Reason for no coverage	Hint
(count == 0)		Y		
Rows:		Hits	FEC Target	Non-masking condition(s)
Row	1:	1	(count == 0)_0	-
Row	2:	1	(count == 0)_1	-

-----Focused Condition View-----

Line 69 Item 1 (count == 1)
Condition totals: 1 of 1 input term covered = 100.00%

Input Term		Covered	Reason for no coverage	Hint
(count == 1)		Y		
Rows:		Hits	FEC Target	Non-masking condition(s)
Row	1:	1	(count == 1)_0	-
Row	2:	1	(count == 1)_1	-

-----Focused Condition View-----

Line 71 Item 1 (count == (fifoif.FIFO_DEPTH - 1))
Condition totals: 1 of 1 input term covered = 100.00%

Input Term		Covered	Reason for no coverage	Hint
(count == (fifoif.FIFO_DEPTH - 1))		Y		
Rows:		Hits	FEC Target	Non-masking condition(s)
Row	1:	1	(count == (fifoif.FIFO_DEPTH - 1))_0	-
Row	2:	1	(count == (fifoif.FIFO_DEPTH - 1))_1	-

-----Focused Condition View-----

Line 106 Item 1 (count == fifoif.FIFO_DEPTH)
Condition totals: 1 of 1 input term covered = 100.00%

Input Term		Covered	Reason for no coverage	Hint
(count == fifoif.FIFO_DEPTH)		Y		
Rows:		Hits	FEC Target	Non-masking condition(s)
Row	1:	1	(count == fifoif.FIFO_DEPTH)_0	-
Row	2:	1	(count == fifoif.FIFO_DEPTH)_1	-

-----Focused Condition View-----

Line 113 Item 1 (count == 0)
Condition totals: 1 of 1 input term covered = 100.00%

Input Term		Covered	Reason for no coverage	Hint
(count == 0)		Y		
Rows:		Hits	FEC Target	Non-masking condition(s)
Row	1:	1	(count == 0)_0	-
Row	2:	1	(count == 0)_1	-

```

-----Focused Condition View-----
Line      120 Item      1  (count == (fifoif.FIFO_DEPTH - 1))
Condition totals: 1 of 1 input term covered = 100.00%

```

	Input Term	Covered	Reason for no coverage	Hint
	-----	-----	-----	-----
	(count == (fifoif.FIFO_DEPTH - 1))	Y		
Rows:	Hits	FEC Target	Non-masking	
condition(s)	-----	-----	-----	
Row 1:	1	(count == (fifoif.FIFO_DEPTH - 1))_0	-	
Row 2:	1	(count == (fifoif.FIFO_DEPTH - 1))_1	-	

```

-----Focused Condition View-----
Line      127 Item      1  (count == 1)
Condition totals: 1 of 1 input term covered = 100.00%

```

	Input Term	Covered	Reason for no coverage	Hint
	-----	-----	-----	-----
	(count == 1)	Y		
Rows:	Hits	FEC Target	Non-masking condition(s)	
	-----	-----	-----	
Row 1:	1	(count == 1)_0	-	
Row 2:	1	(count == 1)_1	-	

```

Directive Coverage:
  Directives              9          9          0  100.00%

```

DIRECTIVE COVERAGE:

Name	Design	Design	Lang	File(Line)	Hits
Status	Unit	UnitType			
-----	-----	-----	-----	-----	-----
/FIFO_top/DUT/reset_state_cp	FIFO	Verilog	SVA	FIFO.sv(92)	460
Covered					
/FIFO_top/DUT/full_flag_cp	FIFO	Verilog	SVA	FIFO.sv(108)	830
Covered					
/FIFO_top/DUT/empty_flag_cp	FIFO	Verilog	SVA	FIFO.sv(115)	545
Covered					
/FIFO_top/DUT/almost_full_flag_cp	FIFO	Verilog	SVA	FIFO.sv(122)	1062
Covered					
/FIFO_top/DUT/almost_empty_flag_cp	FIFO	Verilog	SVA	FIFO.sv(129)	627
Covered					
/FIFO_top/DUT/overflow_behavior_cp	FIFO	Verilog	SVA	FIFO.sv(171)	1621
Covered					
/FIFO_top/DUT/underflow_behavior_cp	FIFO	Verilog	SVA	FIFO.sv(172)	237
Covered					
/FIFO_top/DUT/write_ack_behavior_cp	FIFO	Verilog	SVA	FIFO.sv(173)	4806
Covered					
/FIFO_top/DUT/pointer_limit_cp	FIFO	Verilog	SVA	FIFO.sv(174)	9521
Covered					

```

Statement Coverage:
  Enabled Coverage      Bins      Hits      Misses  Coverage

```


----- Statements	----- 35	----- 35	----- 0	----- 100.00%
=====Statement Details=====				
Statement Coverage for instance /FIFO_top/DUT --				
Line ----	Item ----	Count -----	Source -----	
File FIFO.sv				
1			module FIFO(FIFO_if.DUT	
fifoif);				
2				
3			localparam max_fifo_addr =	
\$clog2(fifoif.FIFO_DEPTH);				
4				
5			reg [fifoif.FIFO_WIDTH-1:0]	
mem [fifoif.FIFO_DEPTH-1:0];				
6				
7			reg [max_fifo_addr-1:0]	
wr_ptr, rd_ptr;				
8			reg [max_fifo_addr:0] count;	
9				
10	1	10462	always @(posedge fifoif.clk	
or negedge fifoif.rst_n) begin				
11				
12			if (! fifoif.rst_n)	
begin				
13	1	941	wr_ptr <= 0;	
14	1	941	fifoif.wr_ack <=0;	
//missing after reset				
15	1	941	fifoif.overflow <=	
0; //missing after reset				
16			end	
17			else if (fifoif.wr_en &&	
count < fifoif.FIFO_DEPTH) begin				
18	1	5044	mem[wr_ptr] <=	
fifoif.data_in;				
19	1	5044	fifoif.wr_ack <= 1;	
20	1	5044	wr_ptr <= wr_ptr +	
1;				
21	1	5044	fifoif.overflow <=	
0;				
22			end	
23			else begin	
24	1	4477	fifoif.wr_ack <= 0;	
25			if (fifoif.full &	
fifoif.wr_en)				
26	1	1712		
fifoif.overflow <= 1;				
27			else	
28	1	2765		
fifoif.overflow <= 0;				
29			end	
30			end	
31				
32	1	10462	always @(posedge fifoif.clk	
or negedge fifoif.rst_n) begin				
33			if (! fifoif.rst_n)	
begin				
34	1	941	rd_ptr <= 0;	
35	1	941	fifoif.data_out <=	
0; //missing after reset				

```

36          1          941          fifoif.underflow <=
0; //missing after reset
37          end
38          else if (fifoif.rd_en &&
count != 0) begin
39          1          2623          fifoif.data_out <=
mem[rd_ptr];
40          1          2623          rd_ptr <= rd_ptr +
1;
41          1          2623          fifoif.underflow <=
0;
42          end
43          else begin
44          if (fifoif.empty &
fifoif.rd_en)
45          1          252          fifoif.underflow <= 1;
46          else
47          1          6646          fifoif.underflow <= 0;
48          end
49          end
50
51          1          9205          always @(posedge fifoif.clk
or negedge fifoif.rst_n) begin
52          if (!fifoif.rst_n) begin
53          1          934          count <= 0;
54          end
55          else begin
56          if (
({fifoif.wr_en, fifoif.rd_en} == 2'b10) && ! fifoif.full)
57          1          3509          count <=
count + 1;
58          else if (
({fifoif.wr_en, fifoif.rd_en} == 2'b01) && ! fifoif.empty)
59          1          766          count <=
count - 1;
60          else if ( ({
fifoif.wr_en,  fifoif.rd_en} == 2'b11) && fifoif.full) //missing cade
61          1          502          count <=
count - 1;
62          else if ( ({
fifoif.wr_en,  fifoif.rd_en} == 2'b11) && count==0) //missing case
63          1          180          count <=
count + 1;
64          end
65          end
66
67          1          5406          assign fifoif.full = (count
== fifoif.FIFO_DEPTH)? 1 : 0;
68          1          5406          assign fifoif.empty = (count
== 0)? 1 : 0;
69          1          5406          assign fifoif.almostempty =
(count == 1)? 1 : 0;
70          //assign fifoif.underflow =
(fifoif.empty && fifoif.rd_en)? 1 : 0; // its mentioned in pdf its a
sequential
71          1          5406          assign fifoif.almostfull =
(count == fifoif.FIFO_DEPTH-1)? 1 : 0; // bug in -2
72
73
74
75          //assertions

```



```

120                                     if(count ==
fifoif.FIFO_DEPTH-1) begin
121     almost_full_flag_ap: assert final(fifoif.almostfull == 1);
122     almost_full_flag_cp: cover final(fifoif.almostfull == 1);
123                                     end
124                                     end
125
126             1             5406     always_comb begin

```

Toggle Coverage:

Enabled Coverage	Bins	Hits	Misses	Coverage
Toggles	20	20	0	100.00%

=====Toggle Details=====

Toggle Coverage for instance /FIFO_top/DUT --

	Node	1H->0L	0L->1H
"Coverage"			

	count[3-0]	1	1
100.00	rd_ptr[2-0]	1	1
100.00	wr_ptr[2-0]	1	1
100.00			

Total Node Count = 10
 Toggled Node Count = 10
 Untoggled Node Count = 0

Toggle Coverage = 100.00% (20 of 20 bins)

=== Instance: /FIFO_top/TEST
 === Design Unit: work.FIFO_tb

Assertion Coverage:

Assertions	1	1	0	100.00%
Name	File(Line)	Failure Count	Pass Count	
/FIFO_top/TEST/#ublk#182146786#18/immed__19	FIFO_tb.sv(19)	0	1	

Statement Coverage:

Enabled Coverage	Bins	Hits	Misses	Coverage
Statements	17	17	0	100.00%

=====Statement Details=====

Statement Coverage for instance /FIFO_top/TEST --

Line	Item	Count	Source
File FIFO_tb.sv			

```

5
fifoif);
6
7
8          1
9          1
10         1
11         1
12         1
13         1
14         1
15         1
16
17         1          10000
18
19
assert(obj_transss.randomize());
20         1          10000
obj_transss.rst_n;
21         1          10000
obj_transss.data_in;
22         1          10000
obj_transss.wr_en;
23         1          10000
obj_transss.rd_en;
24         1          10000
25         1          10000
26
27
28         1          1
29         1          1

```

```

module FIFO_tb(FIFO_if.TEST
FIFO_transaction obj_transss ;
initial begin
obj_transss=new();
    fifoif.rst_n=0;
    fifoif.wr_en=0;
    fifoif.rd_en=0;
    fifoif.data_in=0;
    -> trigger;
    @(negedge fifoif.clk);
    fifoif.rst_n=1;

```

```

repeat(10000)
begin

```

```

    fifoif.rst_n =
    fifoif.data_in =
    fifoif.wr_en =
    fifoif.rd_en =
    -> trigger;
    @(negedge fifoif.clk);
end

```

```

    test_finished=1;
    -> trigger;

```

```

==== Instance: /FIFO_top/MONITOR
==== Design Unit: work.FIFO_monitor
=====

```

Branch Coverage:

Enabled Coverage	Bins	Hits	Misses	Coverage
Branches	2	2	0	100.00%

====Branch Details=====

Branch Coverage for instance /FIFO_top/MONITOR

Line	Item	Count	Source
File monitor.sv			
-----IF Branch-----			
40		10002	Count coming in to IF
40	1	1	if (test_finished==1) begin
		10001	All False Count

Branch totals: 2 hits of 2 branches = 100.00%

Statement Coverage:

Enabled Coverage	Bins	Hits	Misses	Coverage
Statements	23	23	0	100.00%

====Statement Details=====

Statement Coverage for instance /FIFO_top/MONITOR --

Line ----	Item ----	Count -----	Source -----
File monitor.sv			
5			module
FIFO_monitor(FIFO_if.MONITOR fifoif);			
6	1	1	FIFO_transaction
obj_trans=new();			
7	1	1	FIFO_coverage obj_cvr=new();
8	1	1	FIFO_scoreboard obj_sb=new();
9			
10			initial begin
11			
12	1	1	forever begin
13	1	10002	wait(trigger.triggered);
14	1	10002	@(negedge fifoif.clk);
15	1	10002	obj_trans.rst_n =
fifoif.rst_n;			
16	1	10002	obj_trans.wr_en =
fifoif.wr_en;			
17	1	10002	obj_trans.rd_en =
fifoif.rd_en;			
18	1	10002	obj_trans.data_in =
fifoif.data_in;			
19	1	10002	obj_trans.overflow =
fifoif.overflow;			
20	1	10002	obj_trans.underflow =
fifoif.underflow;			
21	1	10002	obj_trans.empty =
fifoif.empty;			
22	1	10002	obj_trans.almostempty =
fifoif.almostempty;			
23	1	10002	obj_trans.full = fifoif.full;
24	1	10002	obj_trans.almostfull =
fifoif.almostfull;			
25	1	10002	obj_trans.data_out =
fifoif.data_out;			
26	1	10002	obj_trans.wr_ack =
fifoif.wr_ack;			
27			
28			fork
29			//first
30			begin
31	1	10002	
obj_cvr.sample_data(obj_trans);			
32			end
33			
34			//second
35			begin
36	1	10002	obj_sb.check_data(obj_trans);
37			end
38			join
39			
40			if (test_finished==1) begin
41	1	1	\$display("Correct count =
%d", correct_count);			
42	1	1	\$display("Error count =
%d", error_count);			
43	1	1	\$stop;

=====

```

=== Instance: /FIFO_top
=== Design Unit: work.FIFO_top
=====

```

Assertion Coverage:

Assertions	1	1	0	100.00%

Name	File(Line)	Failure Count	Pass Count	

/FIFO_top/reset_assert_top	FIFO_top.sv(14)	0	1	

Branch Coverage:

Enabled Coverage	Bins	Hits	Misses	Coverage
-----	----	----	-----	-----
Branches	2	2	0	100.00%

=====Branch Details=====

Branch Coverage for instance /FIFO_top

NOTE: The modification timestamp for source file 'FIFO_top.sv' has been altered since compilation.

Line	Item	Count	Source
----	----	-----	-----
File FIFO_top.sv			
-----IF Branch-----			
13		7812	Count coming in to IF
		6903	All False Count
Branch totals: 2 hits of 2 branches = 100.00%			

Directive Coverage:

Directives	1	1	0	100.00%
------------	---	---	---	---------

DIRECTIVE COVERAGE:

Name	Design	Design	Lang	File(Line)	Hits
Status	Unit	UnitType			

/FIFO_top/reset_cvr_top	FIFO_top	Verilog	SVA	FIFO_top.sv(16)	
460 Covered					

Statement Coverage:

Enabled Coverage	Bins	Hits	Misses	Coverage
-----	----	----	-----	-----
Statements	5	5	0	100.00%

=====Statement Details=====

Statement Coverage for instance /FIFO_top --

NOTE: The modification timestamp for source file 'FIFO_top.sv' has been altered since compilation.

Line	Item	Count	Source
----	----	-----	-----
File FIFO_top.sv			
1			module FIFO_top();
2			
3			bit clk;

```

4
5
6
6
6
7
8
9
10
11
(fifoif);
12
1
7812
endmodule
initial begin
  clk=0;
  forever #1 clk=~clk;
end
FIFO_if fifoif (clk);
FIFO_DUT (fifoif);
FIFO_tb TEST (fifoif);
FIFO_monitor MONITOR

```

Toggle Coverage:

Enabled Coverage	Bins	Hits	Misses	Coverage
-----	----	----	-----	-----
Toggles	2	2	0	100.00%

=====Toggle Details=====

Toggle Coverage for instance /FIFO_top --

	Node	1H->0L	0L->1H
"Coverage"			

	clk	1	1
100.00			

Total Node Count = 1
Toggled Node Count = 1
Untoggled Node Count = 0

Toggle Coverage = 100.00% (2 of 2 bins)

==== Instance: /FIFO_transaction_pkg
=== Design Unit: work.FIFO_transaction_pkg
=====

Statement Coverage:

Enabled Coverage	Bins	Hits	Misses	Coverage
-----	----	----	-----	-----
Statements	2	2	0	100.00%

=====Statement Details=====

Statement Coverage for instance /FIFO_transaction_pkg --

Line	Item	Count	Source
----	----	-----	-----
	File transaction_pkg.sv		
1			package FIFO_transaction_pkg;
2			parameter FIFO_WIDTH =16;
3			parameter FIFO_DEPTH =8;
4			class FIFO_transaction;
5			rand logic wr_en, rd_en,
rst_n;			
6			rand logic [FIFO_WIDTH-1:0]
data_in;			
7			logic [FIFO_WIDTH-1:0]
data_out;			
8			logic wr_ack, overflow;


```

9
almostfull, almostempty, underflow;
10
11
12
dist {1:=95, 0:=5};}
13
dist {1:=WR_EN_ON_DIST, 0:=(100-WR_EN_ON_DIST)};}
14
dist {1:=RD_EN_ON_DIST, 0:=(100-RD_EN_ON_DIST)};}
15
16
RD_EN_ON_DIST=30, int WR_EN_ON_DIST = 70);
17          1          3
RD_EN_ON_DIST;
18          1          3
WR_EN_ON_DIST;

```

```

logic full, empty,

int RD_EN_ON_DIST;
int WR_EN_ON_DIST;
constraint rst_n_dist { rst_n

constraint wr_en_dist { wr_en

constraint rd_en_dist { rd_en

function new(int

this.RD_EN_ON_DIST=

this.WR_EN_ON_DIST=

```

```

=====
=== Instance: /FIFO_coverage_pkg
=== Design Unit: work.FIFO_coverage_pkg
=====

```

Covergroup Coverage:

Covergroups	1	na	na	100.00%
Coverpoints/Crosses	16	na	na	na
Covergroup Bins	66	66	0	100.00%

Covergroup	Metric	Goal
Bins Status		
TYPE /FIFO_coverage_pkg/FIFO_coverage/FIFOcg	100.00%	100
- Covered		
covered/total bins:	66	66
- missing/total bins:	0	66
- % Hit:	100.00%	100
- Coverpoint wr_en_cp	100.00%	100
- Covered		
covered/total bins:	2	2
- missing/total bins:	0	2
- % Hit:	100.00%	100
- bin auto[0]	2908	1
- Covered		
bin auto[1]	7094	1
- Covered		
Coverpoint rd_en_cp	100.00%	100
- Covered		
covered/total bins:	2	2
- missing/total bins:	0	2
- % Hit:	100.00%	100

-	bin auto[0]	6972	1
-	Covered		
-	bin auto[1]	3030	1
-	Covered		
-	Coverpoint full_cp	100.00%	100
-	Covered		
-	covered/total bins:	2	2
-			
-	missing/total bins:	0	2
-			
-	% Hit:	100.00%	100
-			
-	bin auto[0]	7503	1
-	Covered		
-	bin auto[1]	2499	1
-	Covered		
-	Coverpoint almostfull_cp	100.00%	100
-	Covered		
-	covered/total bins:	2	2
-			
-	missing/total bins:	0	2
-			
-	% Hit:	100.00%	100
-			
-	bin auto[0]	8270	1
-	Covered		
-	bin auto[1]	1732	1
-	Covered		
-	Coverpoint empty_cp	100.00%	100
-	Covered		
-	covered/total bins:	2	2
-			
-	missing/total bins:	0	2
-			
-	% Hit:	100.00%	100
-			
-	bin auto[0]	9223	1
-	Covered		
-	bin auto[1]	779	1
-	Covered		
-	Coverpoint almostempty_cp	100.00%	100
-	Covered		
-	covered/total bins:	2	2
-			
-	missing/total bins:	0	2
-			
-	% Hit:	100.00%	100
-			
-	bin auto[0]	8978	1
-	Covered		
-	bin auto[1]	1024	1
-	Covered		
-	Coverpoint overflow_cp	100.00%	100
-	Covered		
-	covered/total bins:	2	2
-			
-	missing/total bins:	0	2
-			
-	% Hit:	100.00%	100
-			
-	bin auto[0]	8290	1
-	Covered		

-	bin auto[1]	1712	1
-	Covered		
-	Coverpoint underflow_cp	100.00%	100
-	Covered		
-	covered/total bins:	2	2
-			
-	missing/total bins:	0	2
-			
-	% Hit:	100.00%	100
-			
-	bin auto[0]	9750	1
-	Covered		
-	bin auto[1]	252	1
-	Covered		
-	Coverpoint wr_ack_cp	100.00%	100
-	Covered		
-	covered/total bins:	2	2
-			
-	missing/total bins:	0	2
-			
-	% Hit:	100.00%	100
-			
-	bin auto[0]	4958	1
-	Covered		
-	bin auto[1]	5044	1
-	Covered		
-	Cross full_cross	100.00%	100
-	Covered		
-	covered/total bins:	6	6
-			
-	missing/total bins:	0	6
-			
-	% Hit:	100.00%	100
-			
-	Auto, Default and User Defined Bins:		
-	bin <auto[1],auto[1],auto[0]>	2151	1
-	Covered		
-	bin <auto[0],auto[1],auto[0]>	879	1
-	Covered		
-	bin <auto[1],auto[0],auto[1]>	2040	1
-	Covered		
-	bin <auto[1],auto[0],auto[0]>	2903	1
-	Covered		
-	bin <auto[0],auto[0],auto[1]>	459	1
-	Covered		
-	bin <auto[0],auto[0],auto[0]>	1570	1
-	Covered		
-	Illegal and Ignore Bins:		
-	ignore_bin rd_full	0	
-	ZERO		
-	Cross almostfull_cross	100.00%	100
-	Covered		
-	covered/total bins:	8	8
-			
-	missing/total bins:	0	8
-			
-	% Hit:	100.00%	100
-			
-	Auto, Default and User Defined Bins:		
-	bin <auto[1],auto[1],auto[1]>	827	1
-	Covered		
-	bin <auto[0],auto[1],auto[1]>	201	1
-	Covered		

-	Covered	bin <auto[1],auto[0],auto[1]>	359	1
-	Covered	bin <auto[0],auto[0],auto[1]>	345	1
-	Covered	bin <auto[1],auto[1],auto[0]>	1324	1
-	Covered	bin <auto[0],auto[1],auto[0]>	678	1
-	Covered	bin <auto[1],auto[0],auto[0]>	4584	1
-	Covered	bin <auto[0],auto[0],auto[0]>	1684	1
-	Covered	Cross empty_cross	100.00%	100
-	Covered	covered/total bins:	8	8
-		missing/total bins:	0	8
-		% Hit:	100.00%	100
-		Auto, Default and User Defined Bins:		
-	Covered	bin <auto[1],auto[1],auto[1]>	114	1
-	Covered	bin <auto[0],auto[1],auto[1]>	210	1
-	Covered	bin <auto[1],auto[0],auto[1]>	224	1
-	Covered	bin <auto[0],auto[0],auto[1]>	231	1
-	Covered	bin <auto[1],auto[1],auto[0]>	2037	1
-	Covered	bin <auto[0],auto[1],auto[0]>	669	1
-	Covered	bin <auto[1],auto[0],auto[0]>	4719	1
-	Covered	bin <auto[0],auto[0],auto[0]>	1798	1
-	Covered	Cross almostempty_cross	100.00%	100
-	Covered	covered/total bins:	8	8
-		missing/total bins:	0	8
-		% Hit:	100.00%	100
-		Auto, Default and User Defined Bins:		
-	Covered	bin <auto[1],auto[1],auto[1]>	373	1
-	Covered	bin <auto[0],auto[1],auto[1]>	82	1
-	Covered	bin <auto[1],auto[0],auto[1]>	365	1
-	Covered	bin <auto[0],auto[0],auto[1]>	204	1
-	Covered	bin <auto[1],auto[1],auto[0]>	1778	1
-	Covered	bin <auto[0],auto[1],auto[0]>	797	1
-	Covered	bin <auto[1],auto[0],auto[0]>	4578	1
-	Covered	bin <auto[0],auto[0],auto[0]>	1825	1
-	Covered			

	Cross overflow_cross	100.00%	100
-	Covered		
	covered/total bins:	6	6
-			
	missing/total bins:	0	6
-			
	% Hit:	100.00%	100
-			
	Auto, Default and User Defined Bins:		
	bin <auto[1],auto[1],auto[1]>	502	1
-	Covered		
	bin <auto[1],auto[0],auto[1]>	1210	1
-	Covered		
	bin <auto[1],auto[1],auto[0]>	1649	1
-	Covered		
	bin <auto[0],auto[1],auto[0]>	879	1
-	Covered		
	bin <auto[1],auto[0],auto[0]>	3733	1
-	Covered		
	bin <auto[0],auto[0],auto[0]>	2029	1
-	Covered		
	Illegal and Ignore Bins:		
	ignore_bin not_wr_overflow	0	
-	ZERO		
	Cross underflow_cross	100.00%	100
-	Covered		
	covered/total bins:	6	6
-			
	missing/total bins:	0	6
-			
	% Hit:	100.00%	100
-			
	Auto, Default and User Defined Bins:		
	bin <auto[1],auto[1],auto[1]>	180	1
-	Covered		
	bin <auto[1],auto[1],auto[0]>	1971	1
-	Covered		
	bin <auto[0],auto[1],auto[1]>	72	1
-	Covered		
	bin <auto[0],auto[1],auto[0]>	807	1
-	Covered		
	bin <auto[1],auto[0],auto[0]>	4943	1
-	Covered		
	bin <auto[0],auto[0],auto[0]>	2029	1
-	Covered		
	Illegal and Ignore Bins:		
	ignore_bin not_rd_overflow	0	
-	ZERO		
	Cross wr_ack_cross	100.00%	100
-	Covered		
	covered/total bins:	6	6
-			
	missing/total bins:	0	6
-			
	% Hit:	100.00%	100
-			
	Auto, Default and User Defined Bins:		
	bin <auto[1],auto[1],auto[1]>	1535	1
-	Covered		
	bin <auto[1],auto[0],auto[1]>	3509	1
-	Covered		
	bin <auto[1],auto[1],auto[0]>	616	1
-	Covered		

```

-      bin <auto[0],auto[1],auto[0]>      879      1
-      Covered
-      bin <auto[1],auto[0],auto[0]>      1434      1
-      Covered
-      bin <auto[0],auto[0],auto[0]>      2029      1
-      Covered
      Illegal and Ignore Bins:
      ignore_bin not_wren_wrack      0
-      ZERO

```

Statement Coverage:

Enabled Coverage	Bins	Hits	Misses	Coverage
-----	----	----	-----	-----
Statements	4	4	0	100.00%

=====Statement Details=====

Statement Coverage for instance /FIFO_coverage_pkg --

Line	Item	Count	Source
----	----	-----	-----
File coverage_pkg.sv			
1			package FIFO_coverage_pkg;
2			import
FIFO_transaction_pkg::* ;			
3			class FIFO_coverage;
4	1	1	FIFO_transaction F_cvg_txn =
new();			
5			
6			covergroup FIFOcg ;
7			
8			wr_en_cp: coverpoint
F_cvg_txn.wr_en;			
9			rd_en_cp: coverpoint
F_cvg_txn.rd_en;			
10			full_cp: coverpoint
F_cvg_txn.full;			
11			almostfull_cp: coverpoint
F_cvg_txn.almostfull;			
12			empty_cp: coverpoint
F_cvg_txn.empty;			
13			almostempty_cp: coverpoint
F_cvg_txn.almostempty;			
14			overflow_cp: coverpoint
F_cvg_txn.overflow;			
15			underflow_cp: coverpoint
F_cvg_txn.underflow;			
16			wr_ack_cp: coverpoint
F_cvg_txn.wr_ack;			
17			
18			//ignored when rd_en =1 with
full not important			
19			full_cross: cross wr_en_cp,
rd_en_cp, full_cp{			
20			ignore_bins rd_full =
binsof(rd_en_cp)intersect{1} &&			
21			binsof(full_cp)intersect{1}
22			;}
23			almostfull_cross: cross
wr_en_cp, rd_en_cp, almostfull_cp;			
24			empty_cross: cross wr_en_cp,
rd_en_cp, empty_cp;			
25			almostempty_cross: cross
wr_en_cp, rd_en_cp, almostempty_cp;			

```

26
27 //ignored when wr_en =0 and
overflow =1 not important
28 overflow_cross: cross
wr_en_cp, rd_en_cp, overflow_cp{
29 ignore_bins
not_wr_overflow = binsof(wr_en_cp)intersect{0} &&
30
binsof(overflow_cp)intersect{1}
31 ;}
32
33 //ignored when rd_en =0 and
underflow =1 not important
34 underflow_cross: cross
wr_en_cp, rd_en_cp, underflow_cp{
35 ignore_bins
not_rd_overflow = binsof(rd_en_cp)intersect{0} &&
36
binsof(underflow_cp)intersect{1}
37 ;}
38
39 //ignored when wr_en=0 with
wr_ack =1 not important
40 wr_ack_cross: cross wr_en_cp,
rd_en_cp, wr_ack_cp{
41 ignore_bins
not_wren_wrack = binsof(wr_en_cp)intersect{0} &&
42 binsof(wr_ack_cp)intersect{1}
43 ;}
44
45 endgroup
46
47 function new();
48 1 FIFOcg =new();
49 endfunction
50
51 function void
sample_data(input FIFO_transaction F_txn);
52 1 10002 F_cvg_txn=F_txn;
53 1 10002 FIFOcg.sample();

```

```

=====
=== Instance: /FIFO_scoreboard_pkg
=== Design Unit: work.FIFO_scoreboard_pkg
=====

```

Branch Coverage:

Enabled Coverage	Bins	Hits	Misses	Coverage
-----	----	----	-----	-----
Branches	27	26	1	96.29%

=====Branch Details=====

Branch Coverage for instance /FIFO_scoreboard_pkg

Line	Item	Count	Source
----	----	-----	-----
File scoreboard_pkg.sv			
-----IF Branch-----			
15		10002	Count coming in to IF
15	1	10002	if (data_out_ref ==
fifotrans_obj_checkdata.data_out && full_ref == fifotrans_obj_checkdata.full &&			
almostfull_ref == fifotrans_obj_checkdata.almostfull && empty_ref ==			

```

fifotrans_obj_checkdata.empty && almostempty_ref ==
fifotrans_obj_checkdata.almostempty && overflow_ref ==
fifotrans_obj_checkdata.overflow && underflow_ref ==
fifotrans_obj_checkdata.underflow && wr_ack_ref == fifotrans_obj_checkdata.wr_ack
) begin
    19                                1                ***0***           else begin
Branch totals: 1 hit of 2 branches = 50.00%

-----IF Branch-----
    49                                10002           Count coming in to IF
    49                                481             if (!
fifotrans_obj_refmodel.rst_n) begin
    55                                5044             else if
(fifotrans_obj_refmodel.wr_en && count < FIFO_DEPTH) begin
    62                                4477             else begin
Branch totals: 3 hits of 3 branches = 100.00%

-----IF Branch-----
    64                                4477           Count coming in to IF
    64                                1712           if (full_ref &&
fifotrans_obj_refmodel.wr_en)
    66                                2765           else
Branch totals: 2 hits of 2 branches = 100.00%

-----IF Branch-----
    71                                10002           Count coming in to IF
    71                                481             if
(!fifotrans_obj_refmodel.rst_n) begin
    77                                2623           else if
(fifotrans_obj_refmodel.rd_en && !empty_ref) begin
    83                                6898           else begin
Branch totals: 3 hits of 3 branches = 100.00%

-----IF Branch-----
    84                                6898           Count coming in to IF
    84                                252             if (
fifotrans_obj_refmodel.rd_en && empty_ref)
    86                                6646           else
Branch totals: 2 hits of 2 branches = 100.00%

-----IF Branch-----
    90                                10002           Count coming in to IF
    90                                481             if
(!fifotrans_obj_refmodel.rst_n) begin
    93                                9521           else begin
Branch totals: 2 hits of 2 branches = 100.00%

-----IF Branch-----
    94                                9521           Count coming in to IF
    94                                3509           if (
({fifotrans_obj_refmodel.wr_en, fifotrans_obj_refmodel.rd_en} == 2'b10) && !
full_ref)
    96                                766             else if (
({fifotrans_obj_refmodel.wr_en, fifotrans_obj_refmodel.rd_en} == 2'b01) && !
empty_ref)
    98                                502             else if ( ({
fifotrans_obj_refmodel.wr_en, fifotrans_obj_refmodel.rd_en} == 2'b11) &&
full_ref) //read ->decrement
    100                               180             else if ( ({
fifotrans_obj_refmodel.wr_en, fifotrans_obj_refmodel.rd_en} == 2'b11) &&
empty_ref) //write ->increment
                                4564           All False Count
Branch totals: 5 hits of 5 branches = 100.00%

```



```

-----IF Branch-----
      104                                10002    Count coming in to IF
      104                                2499      full_ref = (count ==
FIFO_DEPTH)? 1 : 0;
      104                                7503      full_ref = (count ==
FIFO_DEPTH)? 1 : 0;
Branch totals: 2 hits of 2 branches = 100.00%

```

```

-----IF Branch-----
      105                                10002    Count coming in to IF
      105                                779      empty_ref = (count ==
0)? 1 : 0;
      105                                9223      empty_ref = (count ==
0)? 1 : 0;
Branch totals: 2 hits of 2 branches = 100.00%

```

```

-----IF Branch-----
      106                                10002    Count coming in to IF
      106                                1732      almostfull_ref = (count
== FIFO_DEPTH-1)? 1 : 0;
      106                                8270      almostfull_ref = (count
== FIFO_DEPTH-1)? 1 : 0;
Branch totals: 2 hits of 2 branches = 100.00%

```

```

-----IF Branch-----
      107                                10002    Count coming in to IF
      107                                1024      almostempty_ref = (count
== 1)? 1 : 0;
      107                                8978      almostempty_ref = (count
== 1)? 1 : 0;
Branch totals: 2 hits of 2 branches = 100.00%

```

Condition Coverage:

Enabled Coverage	Bins	Covered	Misses	Coverage
Conditions	32	23	9	71.87%

====Condition Details=====

Condition Coverage for instance /FIFO_scoreboard_pkg --

File scoreboard_pkg.sv

-----Focused Condition View-----

```

Line      15 Item      1 ((this.data_out_ref == fifotrans_obj_checkdata.data_out)
&& (this.full_ref == fifotrans_obj_checkdata.full) && (this.almostfull_ref ==
fifotrans_obj_checkdata.almostfull) && (this.empty_ref ==
fifotrans_obj_checkdata.empty) && (this.almostempty_ref ==
fifotrans_obj_checkdata.almostempty) && (this.overflow_ref ==
fifotrans_obj_checkdata.overflow) && (this.underflow_ref ==
fifotrans_obj_checkdata.underflow) && (this.wr_ack_ref ==
fifotrans_obj_checkdata.wr_ack))
Condition totals: 0 of 8 input terms covered = 0.00%

```

	Input Term	Covered	Reason
for no coverage	Hint		
	(this.data_out_ref == fifotrans_obj_checkdata.data_out)	N	'_0'
not hit	Hit '_0'		
	(this.full_ref == fifotrans_obj_checkdata.full)	N	'_0'
not hit	Hit '_0'		

```

        (this.almostfull_ref == fifotrans_obj_checkdata.almostfull)      N   '_0'
not hit                               Hit '_0'
        (this.empty_ref == fifotrans_obj_checkdata.empty)                N   '_0'
not hit                               Hit '_0'
        (this.almostempty_ref == fifotrans_obj_checkdata.almostempty)    N   '_0'
not hit                               Hit '_0'
        (this.overflow_ref == fifotrans_obj_checkdata.overflow)          N   '_0'
not hit                               Hit '_0'
        (this.underflow_ref == fifotrans_obj_checkdata.underflow)        N   '_0'
not hit                               Hit '_0'
        (this.wr_ack_ref == fifotrans_obj_checkdata.wr_ack)              N   '_0'
not hit                               Hit '_0'

```

```

      Rows:      Hits  FEC Target
Non-masking condition(s)
-----

```

```

Row 1:      ***0*** (this.data_out_ref == fifotrans_obj_checkdata.data_out)_0
-
Row 2:      1 (this.data_out_ref == fifotrans_obj_checkdata.data_out)_1
((this.full_ref == fifotrans_obj_checkdata.full) && ((this.almostfull_ref ==
fifotrans_obj_checkdata.almostfull) && ((this.empty_ref ==
fifotrans_obj_checkdata.empty) && ((this.almostempty_ref ==
fifotrans_obj_checkdata.almostempty) && ((this.overflow_ref ==
fifotrans_obj_checkdata.overflow) && ((this.underflow_ref ==
fifotrans_obj_checkdata.underflow) && (this.wr_ack_ref ==
fifotrans_obj_checkdata.wr_ack))))))
Row 3:      ***0*** (this.full_ref == fifotrans_obj_checkdata.full)_0
(this.data_out_ref == fifotrans_obj_checkdata.data_out)
Row 4:      1 (this.full_ref == fifotrans_obj_checkdata.full)_1
((this.data_out_ref == fifotrans_obj_checkdata.data_out) && ((this.almostfull_ref
== fifotrans_obj_checkdata.almostfull) && ((this.empty_ref ==
fifotrans_obj_checkdata.empty) && ((this.almostempty_ref ==
fifotrans_obj_checkdata.almostempty) && ((this.overflow_ref ==
fifotrans_obj_checkdata.overflow) && ((this.underflow_ref ==
fifotrans_obj_checkdata.underflow) && (this.wr_ack_ref ==
fifotrans_obj_checkdata.wr_ack))))))
Row 5:      ***0*** (this.almostfull_ref ==
fifotrans_obj_checkdata.almostfull)_0 ((this.data_out_ref ==
fifotrans_obj_checkdata.data_out) && (this.full_ref ==
fifotrans_obj_checkdata.full))
Row 6:      1 (this.almostfull_ref ==
fifotrans_obj_checkdata.almostfull)_1 ((this.data_out_ref ==
fifotrans_obj_checkdata.data_out) && (this.full_ref ==
fifotrans_obj_checkdata.full) && ((this.empty_ref ==
fifotrans_obj_checkdata.empty) && ((this.almostempty_ref ==
fifotrans_obj_checkdata.almostempty) && ((this.overflow_ref ==
fifotrans_obj_checkdata.overflow) && ((this.underflow_ref ==
fifotrans_obj_checkdata.underflow) && (this.wr_ack_ref ==
fifotrans_obj_checkdata.wr_ack))))))
Row 7:      ***0*** (this.empty_ref == fifotrans_obj_checkdata.empty)_0
((this.data_out_ref == fifotrans_obj_checkdata.data_out) && (this.full_ref ==
fifotrans_obj_checkdata.full) && (this.almostfull_ref ==
fifotrans_obj_checkdata.almostfull))
Row 8:      1 (this.empty_ref == fifotrans_obj_checkdata.empty)_1
((this.data_out_ref == fifotrans_obj_checkdata.data_out) && (this.full_ref ==
fifotrans_obj_checkdata.full) && (this.almostfull_ref ==
fifotrans_obj_checkdata.almostfull) && ((this.almostempty_ref ==
fifotrans_obj_checkdata.almostempty) && ((this.overflow_ref ==
fifotrans_obj_checkdata.overflow) && ((this.underflow_ref ==
fifotrans_obj_checkdata.underflow) && (this.wr_ack_ref ==
fifotrans_obj_checkdata.wr_ack))))))

```

[illegible]

```
Line      55 Item      1  (fifotrans obj refmodel.wr en && (this.count < 8))
```

Condition totals: 2 of 2 input terms covered = 100.00%

	Input Term	Covered	Reason for no coverage	Hint
	-----	-----	-----	-----
	fifotrans_obj_refmodel.wr_en	Y		
	(this.count < 8)	Y		
Rows:	Hits	FEC Target	Non-masking condition(s)	
-----	-----	-----	-----	
Row 1:	1	fifotrans_obj_refmodel.wr_en_0	-	
Row 2:	1	fifotrans_obj_refmodel.wr_en_1	(this.count < 8)	
Row 3:	1	(this.count < 8)_0		
fifotrans_obj_refmodel.wr_en				
Row 4:	1	(this.count < 8)_1		
fifotrans obj_refmodel.wr_en				

-----Focused Condition View-----

Line 64 Item 1 (this.full_ref && fifotrans_obj_refmodel.wr_en)

Condition totals: 2 of 2 input terms covered = 100.00%

		Input Term	Covered	Reason for no coverage	Hint
		-----	-----	-----	-----
		this.full_ref	Y		
		fifotrans_obj_refmodel.wr_en	Y		
Rows:	Hits	FEC Target	Non-masking condition(s)		
-----	-----	-----	-----		
Row 1:	1	this.full_ref_0	-		
Row 2:	1	this.full_ref_1			
		fifotrans_obj_refmodel.wr_en			
Row 3:	1	fifotrans_obj_refmodel.wr_en_0	this.full_ref		
Row 4:	1	fifotrans_obj_refmodel.wr_en_1	this.full_ref		

-----Focused Condition View-----

Line 77 Item 1 (fifotrans_obj_refmodel.rd_en && ~this.empty_ref)

Condition totals: 2 of 2 input terms covered = 100.00%

	Input Term	Covered	Reason for no coverage	Hint
	-----	-----	-----	-----
	fifotrans_obj_refmodel.rd_en	Y		
	this.empty_ref	Y		
Rows:	Hits	FEC Target	Non-masking condition(s)	
-----	-----	-----	-----	
Row 1:	1	fifotrans_obj_refmodel.rd_en_0	-	
Row 2:	1	fifotrans_obj_refmodel.rd_en_1	~this.empty_ref	
Row 3:	1	this.empty_ref_0		
fifotrans_obj_refmodel.rd_en				
Row 4:	1	this.empty_ref_1		
fifotrans obj_refmodel.rd_en				

-----Focused Condition View-----

Line 84 Item 1 (fifotrans_obj_refmodel.rd_en && this.empty_ref)

Condition totals: 1 of 2 input terms covered = 50.00%

		Input Term	Covered	Reason for no coverage	Hint
		-----	-----	-----	-----
fifotrans_obj_refmodel.rd_en			Y		
this.empty_ref			N	'_0' not hit	Hit '_0'
Rows:	Hits	FEC Target	Non-masking condition(s)		
-----	-----	-----	-----		
Row 1:	1	fifotrans_obj_refmodel.rd_en_0	-		

```

Row 2:      1  fifotrans_obj_refmodel.rd_en_1  this.empty_ref
Row 3:      ***0***  this.empty_ref_0
fifotrans_obj_refmodel.rd_en
Row 4:      1  this.empty_ref_1
fifotrans_obj_refmodel.rd_en

```

-----Focused Condition View-----

```

Line      94 Item      1  (~fifotrans_obj_refmodel.rd_en &&
fifotrans_obj_refmodel.wr_en && ~this.full_ref)
Condition totals: 3 of 3 input terms covered = 100.00%

```

Input Term	Covered	Reason for no coverage	Hint
-----	-----	-----	-----
fifotrans_obj_refmodel.rd_en	Y		
fifotrans_obj_refmodel.wr_en	Y		
this.full_ref	Y		

Rows:	Hits	FEC Target	Non-masking condition(s)
-----	-----	-----	-----
Row 1:	1	fifotrans_obj_refmodel.rd_en_0	
(fifotrans_obj_refmodel.wr_en && ~this.full_ref)			
Row 2:	1	fifotrans_obj_refmodel.rd_en_1	-
Row 3:	1	fifotrans_obj_refmodel.wr_en_0	
~fifotrans_obj_refmodel.rd_en			
Row 4:	1	fifotrans_obj_refmodel.wr_en_1	
(~fifotrans_obj_refmodel.rd_en && ~this.full_ref)			
Row 5:	1	this.full_ref_0	
(~fifotrans_obj_refmodel.rd_en && fifotrans_obj_refmodel.wr_en)			
Row 6:	1	this.full_ref_1	
(~fifotrans_obj_refmodel.rd_en && fifotrans_obj_refmodel.wr_en)			

-----Focused Condition View-----

```

Line      96 Item      1  (fifotrans_obj_refmodel.rd_en &&
~fifotrans_obj_refmodel.wr_en && ~this.empty_ref)
Condition totals: 3 of 3 input terms covered = 100.00%

```

Input Term	Covered	Reason for no coverage	Hint
-----	-----	-----	-----
fifotrans_obj_refmodel.rd_en	Y		
fifotrans_obj_refmodel.wr_en	Y		
this.empty_ref	Y		

Rows:	Hits	FEC Target	Non-masking condition(s)
-----	-----	-----	-----
Row 1:	1	fifotrans_obj_refmodel.rd_en_0	-
Row 2:	1	fifotrans_obj_refmodel.rd_en_1	
(~fifotrans_obj_refmodel.wr_en && ~this.empty_ref)			
Row 3:	1	fifotrans_obj_refmodel.wr_en_0	
(fifotrans_obj_refmodel.rd_en && ~this.empty_ref)			
Row 4:	1	fifotrans_obj_refmodel.wr_en_1	
fifotrans_obj_refmodel.rd_en			
Row 5:	1	this.empty_ref_0	
(fifotrans_obj_refmodel.rd_en && ~fifotrans_obj_refmodel.wr_en)			
Row 6:	1	this.empty_ref_1	
(fifotrans_obj_refmodel.rd_en && ~fifotrans_obj_refmodel.wr_en)			

-----Focused Condition View-----

```

Line      98 Item      1  (fifotrans_obj_refmodel.rd_en &&
fifotrans_obj_refmodel.wr_en && this.full_ref)
Condition totals: 3 of 3 input terms covered = 100.00%

```

Input Term	Covered	Reason for no coverage	Hint
-----	-----	-----	-----

fifotrans_obj_refmodel.rd_en	Y
fifotrans_obj_refmodel.wr_en	Y
this.full_ref	Y

Rows:	Hits	FEC Target	Non-masking condition(s)
-----	-----	-----	-----
Row 1:	1	fifotrans_obj_refmodel.rd_en_0	-
Row 2:	1	fifotrans_obj_refmodel.rd_en_1	
(fifotrans_obj_refmodel.wr_en && this.full_ref)			
Row 3:	1	fifotrans_obj_refmodel.wr_en_0	
fifotrans_obj_refmodel.rd_en			
Row 4:	1	fifotrans_obj_refmodel.wr_en_1	
(fifotrans_obj_refmodel.rd_en && this.full_ref)			
Row 5:	1	this.full_ref_0	
(fifotrans_obj_refmodel.rd_en && fifotrans_obj_refmodel.wr_en)			
Row 6:	1	this.full_ref_1	
(fifotrans_obj_refmodel.rd_en && fifotrans_obj_refmodel.wr_en)			

-----Focused Condition View-----
Line 100 Item 1 (fifotrans_obj_refmodel.rd_en && fifotrans_obj_refmodel.wr_en && this.empty_ref)
Condition totals: 3 of 3 input terms covered = 100.00%

Input Term	Covered	Reason for no coverage	Hint
-----	-----	-----	-----
fifotrans_obj_refmodel.rd_en	Y		
fifotrans_obj_refmodel.wr_en	Y		
this.empty_ref	Y		

Rows:	Hits	FEC Target	Non-masking condition(s)
-----	-----	-----	-----
Row 1:	1	fifotrans_obj_refmodel.rd_en_0	-
Row 2:	1	fifotrans_obj_refmodel.rd_en_1	
(fifotrans_obj_refmodel.wr_en && this.empty_ref)			
Row 3:	1	fifotrans_obj_refmodel.wr_en_0	
fifotrans_obj_refmodel.rd_en			
Row 4:	1	fifotrans_obj_refmodel.wr_en_1	
(fifotrans_obj_refmodel.rd_en && this.empty_ref)			
Row 5:	1	this.empty_ref_0	
(fifotrans_obj_refmodel.rd_en && fifotrans_obj_refmodel.wr_en)			
Row 6:	1	this.empty_ref_1	
(fifotrans_obj_refmodel.rd_en && fifotrans_obj_refmodel.wr_en)			

-----Focused Condition View-----
Line 104 Item 1 (this.count == 8)
Condition totals: 1 of 1 input term covered = 100.00%

Input Term	Covered	Reason for no coverage	Hint
-----	-----	-----	-----
(this.count == 8)	Y		

Rows:	Hits	FEC Target	Non-masking condition(s)
-----	-----	-----	-----
Row 1:	1	(this.count == 8)_0	-
Row 2:	1	(this.count == 8)_1	-

-----Focused Condition View-----
Line 105 Item 1 (this.count == 0)
Condition totals: 1 of 1 input term covered = 100.00%

Input Term	Covered	Reason for no coverage	Hint
-----	-----	-----	-----
(this.count == 0)	Y		

Rows:	Hits	FEC Target	Non-masking condition(s)
Row 1:	1	(this.count == 0)_0	-
Row 2:	1	(this.count == 0)_1	-

-----Focused Condition View-----

Line 106 Item 1 (this.count == (8 - 1))
Condition totals: 1 of 1 input term covered = 100.00%

Input Term	Covered	Reason for no coverage	Hint
(this.count == (8 - 1))	Y		

Rows:	Hits	FEC Target	Non-masking condition(s)
Row 1:	1	(this.count == (8 - 1))_0	-
Row 2:	1	(this.count == (8 - 1))_1	-

-----Focused Condition View-----

Line 107 Item 1 (this.count == 1)
Condition totals: 1 of 1 input term covered = 100.00%

Input Term	Covered	Reason for no coverage	Hint
(this.count == 1)	Y		

Rows:	Hits	FEC Target	Non-masking condition(s)
Row 1:	1	(this.count == 1)_0	-
Row 2:	1	(this.count == 1)_1	-

Statement Coverage:

Enabled Coverage	Bins	Hits	Misses	Coverage
Statements	53	30	23	56.60%

=====Statement Details=====

Statement Coverage for instance /FIFO_scoreboard_pkg --

Line	Item	Count	Source
File scoreboard_pkg.sv			
1			package FIFO_scoreboard_pkg;
2			import
FIFO_transaction_pkg::* ;			
3			import shared_pkg::* ;
4			class FIFO_scoreboard;
5			localparam max_fifo_addr =
\$clog2(FIFO_DEPTH);			
6			logic [FIFO_WIDTH-1:0]
data_out_ref;			
7			logic full_ref,
almostfull_ref, empty_ref, almostempty_ref, overflow_ref, underflow_ref,			
wr_ack_ref ;			
8			reg [FIFO_WIDTH-1:0] mem
[FIFO_DEPTH-1:0];			
9			reg [max_fifo_addr-1:0]
wr_ptr, rd_ptr;			
10			reg [max_fifo_addr:0] count;
11			

```

12                                     //Check data
13                                     function void
check_data(input FIFO_transaction fifotrans_obj_checkdata);
14                                     1                                10002
reference_model(fifotrans_obj_checkdata);
15                                     if (data_out_ref ==
fifotrans_obj_checkdata.data_out && full_ref == fifotrans_obj_checkdata.full &&
almostfull_ref == fifotrans_obj_checkdata.almostfull && empty_ref ==
fifotrans_obj_checkdata.empty && almostempty_ref ==
fifotrans_obj_checkdata.almostempty && overflow_ref ==
fifotrans_obj_checkdata.overflow && underflow_ref ==
fifotrans_obj_checkdata.underflow && wr_ack_ref == fifotrans_obj_checkdata.wr_ack
) begin
16                                     1                                10002
correct_count=correct_count+1;
17                                     1                                10002          $display("Correct in
FIFO");
18                                     end
19                                     else begin
20                                     1                                ***0***
error_count=error_count+1;
21                                     1                                ***0***          $display("Error in
FIFO");
22                                     1                                ***0***
$display("rst_n=%0d",fifotrans_obj_checkdata.rst_n);
23                                     1                                ***0***
$display("data_in=%0d",fifotrans_obj_checkdata.data_in);
24                                     1                                ***0***
$display("data_out=%0d",fifotrans_obj_checkdata.data_out);
25                                     1                                ***0***
$display("data_out_ref=%0d",data_out_ref);
26                                     1                                ***0***
$display("wr_ack=%0d",fifotrans_obj_checkdata.wr_ack);
27                                     1                                ***0***
$display("wr_ack_ref=%0d",wr_ack_ref);
28                                     1                                ***0***
$display("overflow=%0d",fifotrans_obj_checkdata.overflow);
29                                     1                                ***0***
$display("overflow_ref=%0d",overflow_ref);
30                                     1                                ***0***
$display("full=%0d",fifotrans_obj_checkdata.full);
31                                     1                                ***0***
$display("full_ref=%0d",full_ref);
32                                     1                                ***0***
$display("empty=%0d",fifotrans_obj_checkdata.empty);
33                                     1                                ***0***
$display("empty_ref=%0d",empty_ref);
34                                     1                                ***0***
$display("almostfull=%0d",fifotrans_obj_checkdata.almostfull);
35                                     1                                ***0***
$display("almostfull_ref=%0d",almostfull_ref);
36                                     1                                ***0***
$display("almostempty=%0d",fifotrans_obj_checkdata.almostempty);
37                                     1                                ***0***
$display("almostempty_ref=%0d",almostempty_ref);
38                                     1                                ***0***
$display("underflow=%0d",fifotrans_obj_checkdata.underflow);
39                                     1                                ***0***
$display("underflow_ref=%0d",underflow_ref);
40                                     1                                ***0***
$display("wr_en=%0d",fifotrans_obj_checkdata.wr_en);
41                                     1                                ***0***
$display("rd_en=%0d",fifotrans_obj_checkdata.rd_en);

```



```

42             1                                     ***0***
$display("count=%0d",count);
43
44                                     end
45                                     endfunction
46
47                                     //Reference model
47                                     function void
reference_model(input FIFO_transaction fifotrans_obj_refmodel);
48                                     //Write data
49                                     if (!
fifotrans_obj_refmodel.rst_n) begin
50                                     1                                     481                                     wr_ptr = 0;
51                                     1                                     481                                     wr_ack_ref = 0;
52                                     1                                     481                                     overflow_ref = 0;
53                                     end
54
55                                     else if
(fifotrans_obj_refmodel.wr_en && count < FIFO_DEPTH) begin
56                                     1                                     5044                                     mem[wr_ptr] =
fifotrans_obj_refmodel.data_in;
57                                     1                                     5044                                     wr_ack_ref = 1;
58                                     1                                     5044                                     wr_ptr = wr_ptr +
1;
59                                     1                                     5044                                     overflow_ref = 0;
60                                     end
61
62                                     else begin
63                                     1                                     4477                                     wr_ack_ref = 0;
64                                     if (full_ref &&
fifotrans_obj_refmodel.wr_en)
65                                     1                                     1712                                     overflow_ref
= 1;
66                                     else
67                                     1                                     2765                                     overflow_ref
= 0;
68                                     end
69
70                                     //Read data
71                                     if
(!fifotrans_obj_refmodel.rst_n) begin
72                                     1                                     481                                     rd_ptr = 0;
73                                     1                                     481                                     underflow_ref = 0;
74                                     1                                     481                                     data_out_ref = 0;
75                                     end
76
77                                     else if
(fifotrans_obj_refmodel.rd_en && !empty_ref) begin
78                                     1                                     2623                                     data_out_ref =
mem[rd_ptr];
79                                     1                                     2623                                     rd_ptr = rd_ptr +
1;
80                                     1                                     2623                                     underflow_ref = 0;
81                                     end
82
83                                     else begin
84                                     if (
fifotrans_obj_refmodel.rd_en && empty_ref)
85                                     1                                     252                                     underflow_ref = 1;
86                                     else
87                                     1                                     6646                                     underflow_ref = 0;
88                                     end
89                                     //Handle count

```

```

90                                     if
(!fifotrans_obj_refmodel.rst_n) begin
91                                     481                                     count = 0;
92                                     end
93                                     else begin
94                                     if (
({fifotrans_obj_refmodel.wr_en, fifotrans_obj_refmodel.rd_en} == 2'b10) && !
full_ref)
95                                     3509                                     count = count
+ 1;
96                                     else if (
({fifotrans_obj_refmodel.wr_en, fifotrans_obj_refmodel.rd_en} == 2'b01) && !
empty_ref)
97                                     766                                     count = count
- 1;
98                                     else if ( ({
fifotrans_obj_refmodel.wr_en, fifotrans_obj_refmodel.rd_en} == 2'b11) &&
full_ref) //read ->decrement
99                                     502                                     count = count
- 1;
100                                    else if ( ({
fifotrans_obj_refmodel.wr_en, fifotrans_obj_refmodel.rd_en} == 2'b11) &&
empty_ref) //write ->increment
101                                    180                                     count = count
+ 1;
102                                    end
103                                    //Handle flags
104                                    10002                                    full_ref = (count ==
FIFO_DEPTH)? 1 : 0;
105                                    10002                                    empty_ref = (count ==
0)? 1 : 0;
106                                    10002                                    almostfull_ref = (count
== FIFO_DEPTH-1)? 1 : 0;
107                                    10002                                    almostempty_ref = (count
== 1)? 1 : 0;

```

COVERGROUP COVERAGE:

Covergroup		Metric	Goal
Bins	Status		

TYPE /FIFO_coverage_pkg/FIFO_coverage/FIFOCg		100.00%	100
- Covered			
covered/total bins:		66	66
- missing/total bins:		0	66
- % Hit:		100.00%	100
- Coverpoint wr_en_cp		100.00%	100
- Covered			
covered/total bins:		2	2
- missing/total bins:		0	2
- % Hit:		100.00%	100
- bin auto[0]		2908	1
- Covered			

-	bin auto[1]	7094	1
-	Covered		
-	Coverpoint rd_en_cp	100.00%	100
-	Covered		
-	covered/total bins:	2	2
-			
-	missing/total bins:	0	2
-			
-	% Hit:	100.00%	100
-			
-	bin auto[0]	6972	1
-	Covered		
-	bin auto[1]	3030	1
-	Covered		
-	Coverpoint full_cp	100.00%	100
-	Covered		
-	covered/total bins:	2	2
-			
-	missing/total bins:	0	2
-			
-	% Hit:	100.00%	100
-			
-	bin auto[0]	7503	1
-	Covered		
-	bin auto[1]	2499	1
-	Covered		
-	Coverpoint almostfull_cp	100.00%	100
-	Covered		
-	covered/total bins:	2	2
-			
-	missing/total bins:	0	2
-			
-	% Hit:	100.00%	100
-			
-	bin auto[0]	8270	1
-	Covered		
-	bin auto[1]	1732	1
-	Covered		
-	Coverpoint empty_cp	100.00%	100
-	Covered		
-	covered/total bins:	2	2
-			
-	missing/total bins:	0	2
-			
-	% Hit:	100.00%	100
-			
-	bin auto[0]	9223	1
-	Covered		
-	bin auto[1]	779	1
-	Covered		
-	Coverpoint almostempty_cp	100.00%	100
-	Covered		
-	covered/total bins:	2	2
-			
-	missing/total bins:	0	2
-			
-	% Hit:	100.00%	100
-			
-	bin auto[0]	8978	1
-	Covered		
-	bin auto[1]	1024	1
-	Covered		

Coverpoint overflow_cp	100.00%	100
- Covered		
covered/total bins:	2	2
-		
missing/total bins:	0	2
-		
% Hit:	100.00%	100
-		
bin auto[0]	8290	1
- Covered		
bin auto[1]	1712	1
- Covered		
Coverpoint underflow_cp	100.00%	100
- Covered		
covered/total bins:	2	2
-		
missing/total bins:	0	2
-		
% Hit:	100.00%	100
-		
bin auto[0]	9750	1
- Covered		
bin auto[1]	252	1
- Covered		
Coverpoint wr_ack_cp	100.00%	100
- Covered		
covered/total bins:	2	2
-		
missing/total bins:	0	2
-		
% Hit:	100.00%	100
-		
bin auto[0]	4958	1
- Covered		
bin auto[1]	5044	1
- Covered		
Cross full_cross	100.00%	100
- Covered		
covered/total bins:	6	6
-		
missing/total bins:	0	6
-		
% Hit:	100.00%	100
-		
Auto, Default and User Defined Bins:		
bin <auto[1],auto[1],auto[0]>	2151	1
- Covered		
bin <auto[0],auto[1],auto[0]>	879	1
- Covered		
bin <auto[1],auto[0],auto[1]>	2040	1
- Covered		
bin <auto[1],auto[0],auto[0]>	2903	1
- Covered		
bin <auto[0],auto[0],auto[1]>	459	1
- Covered		
bin <auto[0],auto[0],auto[0]>	1570	1
- Covered		
Illegal and Ignore Bins:		
ignore_bin rd_full	0	
- ZERO		
Cross almostfull_cross	100.00%	100
- Covered		

-	covered/total bins:	8	8
-	missing/total bins:	0	8
-	% Hit:	100.00%	100
-	Auto, Default and User Defined Bins:		
-	bin <auto[1],auto[1],auto[1]>	827	1
-	Covered bin <auto[0],auto[1],auto[1]>	201	1
-	Covered bin <auto[1],auto[0],auto[1]>	359	1
-	Covered bin <auto[0],auto[0],auto[1]>	345	1
-	Covered bin <auto[1],auto[1],auto[0]>	1324	1
-	Covered bin <auto[0],auto[1],auto[0]>	678	1
-	Covered bin <auto[1],auto[0],auto[0]>	4584	1
-	Covered bin <auto[0],auto[0],auto[0]>	1684	1
-	Covered Cross empty_cross	100.00%	100
-	Covered covered/total bins:	8	8
-	missing/total bins:	0	8
-	% Hit:	100.00%	100
-	Auto, Default and User Defined Bins:		
-	bin <auto[1],auto[1],auto[1]>	114	1
-	Covered bin <auto[0],auto[1],auto[1]>	210	1
-	Covered bin <auto[1],auto[0],auto[1]>	224	1
-	Covered bin <auto[0],auto[0],auto[1]>	231	1
-	Covered bin <auto[1],auto[1],auto[0]>	2037	1
-	Covered bin <auto[0],auto[1],auto[0]>	669	1
-	Covered bin <auto[1],auto[0],auto[0]>	4719	1
-	Covered bin <auto[0],auto[0],auto[0]>	1798	1
-	Covered Cross almostempty_cross	100.00%	100
-	Covered covered/total bins:	8	8
-	missing/total bins:	0	8
-	% Hit:	100.00%	100
-	Auto, Default and User Defined Bins:		
-	bin <auto[1],auto[1],auto[1]>	373	1
-	Covered bin <auto[0],auto[1],auto[1]>	82	1
-	Covered bin <auto[1],auto[0],auto[1]>	365	1
-	Covered		

-	bin <auto[0],auto[0],auto[1]>	204	1
-	Covered		
-	bin <auto[1],auto[1],auto[0]>	1778	1
-	Covered		
-	bin <auto[0],auto[1],auto[0]>	797	1
-	Covered		
-	bin <auto[1],auto[0],auto[0]>	4578	1
-	Covered		
-	bin <auto[0],auto[0],auto[0]>	1825	1
-	Covered		
-	Cross overflow_cross	100.00%	100
-	Covered		
-	covered/total bins:	6	6
-			
-	missing/total bins:	0	6
-			
-	% Hit:	100.00%	100
-			
-	Auto, Default and User Defined Bins:		
-	bin <auto[1],auto[1],auto[1]>	502	1
-	Covered		
-	bin <auto[1],auto[0],auto[1]>	1210	1
-	Covered		
-	bin <auto[1],auto[1],auto[0]>	1649	1
-	Covered		
-	bin <auto[0],auto[1],auto[0]>	879	1
-	Covered		
-	bin <auto[1],auto[0],auto[0]>	3733	1
-	Covered		
-	bin <auto[0],auto[0],auto[0]>	2029	1
-	Covered		
-	Illegal and Ignore Bins:		
-	ignore_bin not_wr_overflow	0	
-	ZERO		
-	Cross underflow_cross	100.00%	100
-	Covered		
-	covered/total bins:	6	6
-			
-	missing/total bins:	0	6
-			
-	% Hit:	100.00%	100
-			
-	Auto, Default and User Defined Bins:		
-	bin <auto[1],auto[1],auto[1]>	180	1
-	Covered		
-	bin <auto[1],auto[1],auto[0]>	1971	1
-	Covered		
-	bin <auto[0],auto[1],auto[1]>	72	1
-	Covered		
-	bin <auto[0],auto[1],auto[0]>	807	1
-	Covered		
-	bin <auto[1],auto[0],auto[0]>	4943	1
-	Covered		
-	bin <auto[0],auto[0],auto[0]>	2029	1
-	Covered		
-	Illegal and Ignore Bins:		
-	ignore_bin not_rd_overflow	0	
-	ZERO		
-	Cross wr_ack_cross	100.00%	100
-	Covered		
-	covered/total bins:	6	6
-			

```

missing/total bins:                                0           6
-
% Hit:                                              100.00%       100
-
Auto, Default and User Defined Bins:
  bin <auto[1],auto[1],auto[1]>                    1535           1
- Covered
  bin <auto[1],auto[0],auto[1]>                    3509           1
- Covered
  bin <auto[1],auto[1],auto[0]>                     616           1
- Covered
  bin <auto[0],auto[1],auto[0]>                     879           1
- Covered
  bin <auto[1],auto[0],auto[0]>                    1434           1
- Covered
  bin <auto[0],auto[0],auto[0]>                    2029           1
- Covered
  Illegal and Ignore Bins:
    ignore_bin not_wren_wrack                      0
- ZERO

```

TOTAL COVERGROUP COVERAGE: 100.00% COVERGROUP TYPES: 1

DIRECTIVE COVERAGE:

Name Status	Design Unit	Design UnitType	Lang	File (Line)	Hits
/FIFO_top/reset_cvr_top 460 Covered	FIFO_top	Verilog	SVA	FIFO_top.sv(16)	
/FIFO_top/DUT/reset_state_cp Covered	FIFO	Verilog	SVA	FIFO.sv(92)	460
/FIFO_top/DUT/full_flag_cp Covered	FIFO	Verilog	SVA	FIFO.sv(108)	830
/FIFO_top/DUT/empty_flag_cp Covered	FIFO	Verilog	SVA	FIFO.sv(115)	545
/FIFO_top/DUT/almost_full_flag_cp Covered	FIFO	Verilog	SVA	FIFO.sv(122)	1062
/FIFO_top/DUT/almost_empty_flag_cp Covered	FIFO	Verilog	SVA	FIFO.sv(129)	627
/FIFO_top/DUT/overflow_behavior_cp Covered	FIFO	Verilog	SVA	FIFO.sv(171)	1621
/FIFO_top/DUT/underflow_behavior_cp Covered	FIFO	Verilog	SVA	FIFO.sv(172)	237
/FIFO_top/DUT/write_ack_behavior_cp Covered	FIFO	Verilog	SVA	FIFO.sv(173)	4806
/FIFO_top/DUT/pointer_limit_cp Covered	FIFO	Verilog	SVA	FIFO.sv(174)	9521

TOTAL DIRECTIVE COVERAGE: 100.00% COVERS: 10

ASSERTION RESULTS:

Name	File (Line)	Failure Count	Pass Count
/FIFO_top/reset_assert_top	FIFO_top.sv(14)	0	1
/FIFO_top/DUT/reset_state_ap	FIFO.sv(82)	0	1

/FIFO_top/DUT/full_flag_ap		
FIFO.sv(107)	0	1
/FIFO_top/DUT/empty_flag_ap		
FIFO.sv(114)	0	1
/FIFO_top/DUT/almost_full_flag_ap		
FIFO.sv(121)	0	1
/FIFO_top/DUT/almost_empty_flag_ap		
FIFO.sv(128)	0	1
/FIFO_top/DUT/overflow_behavior_ap		
FIFO.sv(164)	0	1
/FIFO_top/DUT/underflow_behavior_ap		
FIFO.sv(165)	0	1
/FIFO_top/DUT/write_ack_behavior_ap		
FIFO.sv(166)	0	1
/FIFO_top/DUT/pointer_limit_ap		
FIFO.sv(167)	0	1
/FIFO_top/TEST/#ublk#182146786#18/immed__19		
FIFO_tb.sv(19)	0	1